

Beating the Clock, A Predictive Approach to First Responder Response Time

AUTHOR
Samantha Doyle & Alyssa Holmes

Data Preparation

Data Extraction

This data was sourced from *NYC Open Data*.

```
# install.packages("readxl")
library(readxl)

# Import data
responder <- read.csv('End_to_end_911.csv')
head(responder)
```

Week.Start.Date	Agency	
Final.Incident.Type		
1	02/17/2025 NYPD (Non-CIP)	Police Officer/Security
Holding Suspect		
2	02/17/2025 NYPD (Non-CIP)	Disorderly
Person/Group/Noise		
3	02/17/2025 FDNY	4. Non-Medical
Emergencies		
4	02/17/2025 NYPD (Non-CIP)	
Possible Crimes		
5	02/17/2025 EMS	2. Non-Life Threatening Med
Emergencies		
6	02/17/2025 NYPD (Non-CIP)	Vehicle
Accident		
X..of.Incidents.Calculated Call.to.First.Pickup		
Call.to.PD.Calltaker.Handoff		
1	82	3.65
108.80		
2	1647	3.59
154.82		
3	2315	3.72
136.52		
4	5586	3.70
161.96		

5	10160	3.73
99.54		
6	201	4.21
202.55		
Call.to.FDNY.Pickup	Call.to.FDNY.Job.Creation	
Call.to.EMS.Pickup		
1	NA	NA
NA		
2	NA	NA
NA		
3	54.55	109.5
NA		
4	NA	NA
NA		
5	NA	NA
106.82		
6	NA	NA
NA		
Call.to.Agency.Job.Creation	Call.to.Agency.Dispatch	
Call.to.Agency.Arrival		
1	NA	576.13
1719.06		
2	NA	653.77
3254.60		
3	NA	202.61
546.17		
4	NA	728.33
4006.24		
5	165.44	331.06
991.80		
6	NA	850.13
3394.60		
Call.to.First.Arrival..Multi.Agency.Incidents.	Median.Pickup	
1	1616.93	3
2	3215.73	3
3	496.05	3
4	3856.27	3
5	907.75	3
6	3249.81	3
Median.Calltaker.Handoff	Median.FDNY.Pickup	
Median.FDNY.Job.Creation		
1	75.5	NA
NA		
2	129.0	NA
NA		
3	26.0	5
48		
4	128.0	NA

NA			
5	92.0	NA	
NA			
6	167.0	NA	
NA			
Median.Ems.Pickup	Median.Ems.Job.Creation	Median.Dispatch	
Median.Travel			
1	NA	NA	182.5
514.5			
2	NA	NA	174.0
1168.0			
3	NA	NA	7.0
304.0			
4	NA	NA	213.0
1610.0			
5	11.5	36	99.5
553.0			
6	NA	NA	249.0
1356.0			
Median.Cumulative.First.Arrival...Multi.Agency.Incidents.			
Average.Travel			
1			900.5
19.05			
2			1687.0
43.35			
3			420.0
5.73			
4			2175.0
54.63			
5			689.5
11.01			
6			2028.0
42.41			
Average.Dispatch	Average.EMS.Processing	Average.EMS.Pickup	
1	7.79	NA	NA
2	8.32	NA	NA
3	0.21	NA	NA
4	9.44	NA	NA
5	3.86	1	0.16
6	10.79	NA	NA
Average.Calltaker.Processing	Average.Pickup	Average.FD.Pickup	
1	1.75	0.06	NA
2	2.52	0.06	NA
3	2.21	0.06	0.09
4	2.64	0.06	NA
5	1.60	0.06	NA
6	3.31	0.07	NA
Average.FD.Processing			

1	NA
2	NA
3	0.92
4	NA
5	NA
6	NA

```
attach(responder)
```

Data Cleaning

The original dataset contains one date variable, two categorical variables, and twenty-seven numeric variables. All the numeric variables are measured in seconds except for `X.of.Incidents.Calculated` which counts the number of incidents in a given month. The two categorical variables are `agency`, which states the department of the emergency service, and `Final.Incident.Type`, which states the final incident category. Twelve numeric variables were department specific (i.e. Fire Department) measurements. Since the goal of this model is to predict emergency services as a whole, these variables were removed. A few of the remaining numeric columns had a singular missing value, which was imputed with the mean. A unique ID variable was created for each observation, as well. From the date variable, the month was extracted and sorted into categories by season (three months for spring, summer, fall, and winter each) to obtain another categorical variable. Finally, the missing values in `Final.Incident.Type` were imputed with the mode, and the 23 categories in this variable were condensed into a new categorical variable called `IncidentClassification`.

```
# Remove unwanted variables
responder_clean <- subset(
  responder,
  select = -c(
    Call.to.FDNY.Job.Creation,
    Call.to.FDNY.Pickup,
    Call.to.EMS.Pickup,
    Call.to.Agency.Job.Creation,
    Median.FDNY.Pickup,
    Median.FDNY.Job.Creation,
    Median.Ems.Pickup,
    Median.Ems.Job.Creation,
    Average.EMS.Pickup,
    Average.EMS.Processing,
    Average.FD.Processing,
    Average.FD.Pickup
  )
)
```

```
)
)

head(responder_clean)
```

	Week.Start.Date	Agency	
Final.Incident.Type			
1	02/17/2025	NYPD (Non-CIP)	Police Officer/Security
Holding Suspect			
2	02/17/2025	NYPD (Non-CIP)	Disorderly
Person/Group/Noise			
3	02/17/2025	FDNY	4. Non-Medical
Emergencies			
4	02/17/2025	NYPD (Non-CIP)	
Possible Crimes			
5	02/17/2025	EMS	2. Non-Life Threatening Med
Emergencies			
6	02/17/2025	NYPD (Non-CIP)	Vehicle
Accident			
X..of.Incidents.Calculated	Call.to.First.Pickup		
Call.to.PD.Calltaker.Handoff			
1	82	3.65	
108.80			
2	1647	3.59	
154.82			
3	2315	3.72	
136.52			
4	5586	3.70	
161.96			
5	10160	3.73	
99.54			
6	201	4.21	
202.55			
Call.to.Agency.Dispatch	Call.to.Agency.Arrival		
1	576.13	1719.06	
2	653.77	3254.60	
3	202.61	546.17	
4	728.33	4006.24	
5	331.06	991.80	
6	850.13	3394.60	
Call.to.First.Arrival..Multi.Agency.Incidents.	Median.Pickup		
1	1616.93	3	
2	3215.73	3	
3	496.05	3	
4	3856.27	3	
5	907.75	3	
6	3249.81	3	

	Median.Calltaker.Handoff	Median.Dispatch	Median.Travel
1	75.5	182.5	514.5
2	129.0	174.0	1168.0
3	26.0	7.0	304.0
4	128.0	213.0	1610.0
5	92.0	99.5	553.0
6	167.0	249.0	1356.0

Median.Cumulative.First.Arrival...Multi.Agency.Incidents.
Average.Travel

1	900.5
19.05	
2	1687.0
43.35	
3	420.0
5.73	
4	2175.0
54.63	
5	689.5
11.01	
6	2028.0
42.41	

	Average.Dispatch	Average.Calltaker.Processing	Average.Pickup
1	7.79	1.75	0.06
2	8.32	2.52	0.06
3	0.21	2.21	0.06
4	9.44	2.64	0.06
5	3.86	1.60	0.06
6	10.79	3.31	0.07

```
summary(responder_clean)
```

Week.Start.Date	Agency	Final.Incident.Type
Length:10797	Length:10797	Length:10797
Class :character	Class :character	Class :character
Mode :character	Mode :character	Mode :character

X..of.Incidents.Calculated	Call.to.First.Pickup
Call.to.PD.Calltaker.Handoff	
Min. : 1	Min. : -46.680
-0.84	
1st Qu.: 436	1st Qu.: 3.390
109.78	
Median : 2391	Median : 3.690
149.32	

Mean : 3121	Mean : 3.985	Mean :
142.67		
3rd Qu.: 3814	3rd Qu.: 4.200	3rd Qu.:
172.96		
Max. :17613	Max. : 56.080	Max.
:1459.77		
		NA's :1

Call.to.Agency.Dispatch	Call.to.Agency.Arrival
-------------------------	------------------------

Min. : 70.0	Min. : 110.0
1st Qu.: 214.5	1st Qu.: 544.4
Median : 400.5	Median : 1108.8
Mean : 498.2	Mean : 1700.3
3rd Qu.: 745.2	3rd Qu.: 2663.3
Max. :4331.0	Max. :16395.0

Call.to.First.Arrival..Multi.Agency.Incidents.	Median.Pickup
Min. : 110.0	Min. : 2.000
1st Qu.: 491.6	1st Qu.: 3.000
Median : 1004.5	Median : 3.000
Mean : 1589.4	Mean : 3.047
3rd Qu.: 2516.6	3rd Qu.: 3.000
Max. :16395.0	Max. :27.000

Median.Calltaker.Handoff	Median.Dispatch	Median.Travel
Min. : 12.0	Min. : -950.0	Min. : -452.0
1st Qu.: 81.0	1st Qu.: 57.0	1st Qu.: 267.0
Median :111.0	Median : 105.0	Median : 429.0
Mean :108.1	Mean : 137.6	Mean : 684.9
3rd Qu.:139.5	3rd Qu.: 220.0	3rd Qu.: 969.0
Max. :711.5	Max. :4231.0	Max. :14591.0
NA's :1	NA's :1	

Median.Cumulative.First.Arrival...Multi.Agency.Incidents.	
Average.Travel	
Min. : 110	Min.
: -7.53	
1st Qu.: 402	1st
Qu.: 5.16	
Median : 636	
Median : 11.27	
Mean : 1005	Mean
: 20.04	
3rd Qu.: 1473	3rd
Qu.: 32.24	
Max. :16395	Max.
:243.18	

Average.Dispatch	Average.Calltaker.Processing	Average.Pickup
Min. : -25.920	Min. : -0.070	Min. : -0.7800

1st Qu.:	1.480	1st Qu.:	1.740	1st Qu.:	0.0600
Median :	4.070	Median :	2.420	Median :	0.0600
Mean :	5.814	Mean :	2.305	Mean :	0.0663
3rd Qu.:	9.760	3rd Qu.:	2.810	3rd Qu.:	0.0700
Max. :	70.520	Max. :	24.270	Max. :	0.9300
NA's :	1	NA's :	1		

```
# Find columns with missing values
colSums(is.na(responder_clean))
```

```

Week.Start.Date
0
Agency
0
Final.Incident.Type
0
X..of.Incidents.Calculated
0
Call.to.First.Pickup
0
Call.to.PD.Calltaker.Handoff
1
Call.to.Agency.Dispatch
0
Call.to.Agency.Arrival
0
Call.to.First.Arrival..Multi.Agency.Incidents.
0
Median.Pickup
0
Median.Calltaker.Handoff
1
Median.Dispatch
1
Median.Travel
0
Median.Cumulative.First.Arrival...Multi.Agency.Incidents.
0
Average.Travel
0
Average.Dispatch
1
Average.Calltaker.Processing
1
Average.Pickup
0

```



```

# Replace Average.Calltaker.Processing NA with column mean
responder_clean$Average.Calltaker.Processing[is.na(responder_clean$Average.Calltaker.Processing)]
  <-
  mean(responder_clean$Average.Calltaker.Processing, na.rm =
    TRUE)

# Replace Average.Dispatch NA with column mean
responder_clean$Average.Dispatch[is.na(responder_clean$Average.Dispatch)]
  <-
  mean(responder_clean$Average.Dispatch, na.rm = TRUE)

# Replace Median.Dispatch NA with column mean
responder_clean$Median.Dispatch[is.na(responder_clean$Median.Dispatch)]
  <-
  mean(responder_clean$Median.Dispatch, na.rm = TRUE)

# Replace Median.Calltaker.Handoff NA with column mean
responder_clean$Median.Calltaker.Handoff[is.na(responder_clean$Median.Calltaker.Handoff)]
  <-
  mean(responder_clean$Median.Calltaker.Handoff, na.rm = TRUE)

# Replace Call.to.PD.Calltaker.Handoff NA with column mean
responder_clean$Call.to.PD.Calltaker.Handoff[is.na(responder_clean$Call.to.PD.Calltaker.Handoff)]
  <-
  mean(responder_clean$Call.to.PD.Calltaker.Handoff, na.rm =
    TRUE)

# Recheck missing values
colSums(is.na(responder_clean))

```

```

Week.Start.Date
0
Agency
0
Final.Incident.Type
0
X..of.Incidents.Calculated
0
Call.to.First.Pickup
0
Call.to.PD.Calltaker.Handoff
0
Call.to.Agency.Dispatch
0
Call.to.Agency.Arrival
0
Call.to.First.Arrival..Multi.Agency.Incidents.
0
Median.Pickup

```

```

Median.Calltaker.Handoff
0
Median.Dispatch
0
Median.Travel
0
Median.Cumulative.First.Arrival...Multi.Agency.Incidents.
0
Average.Travel
0
Average.Dispatch
0
Average.Calltaker.Processing
0
Average.Pickup
0

```

```

# Create the ID variable
responder_clean$ID <- 1:nrow(responder_clean)

# Reorder the columns to place 'ID' at the beginning
responder_clean <-
  responder_clean[, c("ID", setdiff(names(responder_clean),
    "ID"))]

# View the first few rows to check the result
head(responder_clean)

```

```

ID Week.Start.Date      Agency
Final.Incident.Type
1  1      02/17/2025 NYPD (Non-CIP) Police Officer/Security
Holding Suspect
2  2      02/17/2025 NYPD (Non-CIP)      Disorderly
Person/Group/Noise
3  3      02/17/2025      FDNY      4. Non-Medical
Emergencies
4  4      02/17/2025 NYPD (Non-CIP)
Possible Crimes
5  5      02/17/2025      EMS 2. Non-Life Threatening Med
Emergencies
6  6      02/17/2025 NYPD (Non-CIP)
Vehicle Accident
X..of.Incidents.Calculated Call.to.First.Pickup
Call.to.PD.Calltaker.Handoff
1      82      3.65
108.80

```

2	1647	3.59
154.82		
3	2315	3.72
136.52		
4	5586	3.70
161.96		
5	10160	3.73
99.54		
6	201	4.21
202.55		

	Call.to.Agency.Dispatch	Call.to.Agency.Arrival
1	576.13	1719.06
2	653.77	3254.60
3	202.61	546.17
4	728.33	4006.24
5	331.06	991.80
6	850.13	3394.60

	Call.to.First.Arrival..Multi.Agency.Incidents.	Median.Pickup
1	1616.93	3
2	3215.73	3
3	496.05	3
4	3856.27	3
5	907.75	3
6	3249.81	3

	Median.Calltaker.Handoff	Median.Dispatch	Median.Travel
1	75.5	182.5	514.5
2	129.0	174.0	1168.0
3	26.0	7.0	304.0
4	128.0	213.0	1610.0
5	92.0	99.5	553.0
6	167.0	249.0	1356.0

	Median.Cumulative.First.Arrival...Multi.Agency.Incidents.
Average.Travel	

1	900.5
19.05	
2	1687.0
43.35	
3	420.0
5.73	
4	2175.0
54.63	
5	689.5
11.01	
6	2028.0
42.41	

	Average.Dispatch	Average.Calltaker.Processing	Average.Pickup
1	7.79	1.75	0.06
2	8.32	2.52	0.06

3	0.21	2.21	0.06
4	9.44	2.64	0.06
5	3.86	1.60	0.06
6	10.79	3.31	0.07

```
# Convert Week.Start.Date to Date format
responder_clean$Week.Start.Date <-
  as.Date(responder_clean$Week.Start.Date, format =
    "%m/%d/%Y")

# Create a new column for Season
responder_clean$Season <- ifelse(
  format(responder_clean$Week.Start.Date, "%m") %in% c("12",
    "01", "02"),
  "Winter",
  ifelse(
    format(responder_clean$Week.Start.Date, "%m") %in% c("03",
      "04", "05"),
    "Spring",
    ifelse(
      format(responder_clean$Week.Start.Date, "%m") %in%
        c("06", "07", "08"),
      "Summer",
      "Fall"
    )
  )
)

# Reorder columns
responder_clean <- responder_clean[, c("ID",
  "Week.Start.Date", "Season",

  setdiff(names(responder_clean),
    c("ID",
      "Week.Start.Date",
      "Season")))]

# View data
head(responder_clean)
```

	ID	Week.Start.Date	Season	Agency
1	1	2025-02-17	Winter	NYPD (Non-CIP)
2	2	2025-02-17	Winter	NYPD (Non-CIP)
3	3	2025-02-17	Winter	FDNY
4	4	2025-02-17	Winter	NYPD (Non-CIP)
5	5	2025-02-17	Winter	EMS
6	6	2025-02-17	Winter	NYPD (Non-CIP)
				Final.Incident.Type
				X..of.Incidents.Calculated

1	Police Officer/Security Holding Suspect		
82			
2	Disorderly Person/Group/Noise		
1647			
3	4. Non-Medical Emergencies		
2315			
4	Possible Crimes		
5586			
5	2. Non-Life Threatening Med Emergencies		
10160			
6	Vehicle Accident		
201			
	Call.to.First.Pickup	Call.to.PD.Calltaker.Handoff	
	Call.to.Agency.Dispatch		
1	3.65	108.80	
576.13			
2	3.59	154.82	
653.77			
3	3.72	136.52	
202.61			
4	3.70	161.96	
728.33			
5	3.73	99.54	
331.06			
6	4.21	202.55	
850.13			
	Call.to.Agency.Arrival		
	Call.to.First.Arrival..Multi.Agency.Incidents.		
1	1719.06		
1616.93			
2	3254.60		
3215.73			
3	546.17		
496.05			
4	4006.24		
3856.27			
5	991.80		
907.75			
6	3394.60		
3249.81			
	Median.Pickup	Median.Calltaker.Handoff	Median.Dispatch
	Median.Travel		
1	3	75.5	182.5
514.5			
2	3	129.0	174.0
1168.0			
3	3	26.0	7.0
304.0			

4	3	128.0	213.0
1610.0			
5	3	92.0	99.5
553.0			
6	3	167.0	249.0
1356.0			

Median.Cumulative.First.Arrival...Multi.Agency.Incidents.
Average.Travel

1	900.5
19.05	
2	1687.0
43.35	
3	420.0
5.73	
4	2175.0
54.63	
5	689.5
11.01	
6	2028.0
42.41	

	Average.Dispatch	Average.Calltaker.Processing	Average.Pickup
1	7.79	1.75	0.06
2	8.32	2.52	0.06
3	0.21	2.21	0.06
4	9.44	2.64	0.06
5	3.86	1.60	0.06
6	10.79	3.31	0.07

```
# Check for empty strings or zero-length entries
colSums(responder_clean == "")
```

ID
0
Week.Start.Date
NA
Season
0
Agency
0
Final.Incident.Type
72
X..of.Incidents.Calculated
0
Call.to.First.Pickup
0
Call.to.PD.Calltaker.Handoff
0

```

Call.to.Agency.Dispatch
0
Call.to.Agency.Arrival
0
Call.to.First.Arrival...Multi.Agency.Incidents.
0
Median.Pickup
0
Median.Calltaker.Handoff
0
Median.Dispatch
0
Median.Travel
0
Median.Cumulative.First.Arrival...Multi.Agency.Incidents.
0
Average.Travel
0
Average.Dispatch
0
Average.Calltaker.Processing
0
Average.Pickup
0

```

```

# Convert empty strings to NA
responder_clean[responder_clean == ""] <- NA

# install.packages("dplyr")
library(dplyr)

```

Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

filter, lag

The following objects are masked from 'package:base':

intersect, setdiff, setequal, union

```

# Reduce factor levels for Final.Incident.Type
responder_clean <- responder_clean %>%
  mutate(
    IncidentClassification = case_when(
      Final.Incident.Type %in% c(

```

```

    "1. Critical",
    "1. Life Threatening Med Emergencies",
    "1. Structural Fires"
  ) ~ "Critical",
  Final.Incident.Type %in% c(
    "2. Non-Life Threatening Med Emergencies",
    "2. Serious",
    "2. Non-Structural Fires"
  ) ~ "Serious",
  Final.Incident.Type %in% c("3. Medical Emergencies", "3.
    Non-Critical") ~
    "Non-Critical",
  Final.Incident.Type %in% c("4. Non-Medical Emergencies")
    ~ "Other",
  Final.Incident.Type %in% c("Alarms", "Shot Spotter") ~
    "Alert",
  Final.Incident.Type %in% c("Disorderly", "Disorderly
    Person/Group/Noise",
    "Dispute") ~ "Dispute",
  Final.Incident.Type %in% c(
    "Hazardous Materials/Suspicious
    Letters/Packages/Substances/Substances",
    "United States Postal Service – Bio Hazard Detection
    System"
  ) ~ "Hazard",
  Final.Incident.Type %in% c(
    "Investigate/Possible Crime",
    "Other Crimes (In Progress)",
    "Possible Crimes"
  ) ~ "Possible Crime",
  Final.Incident.Type %in% c("Past Crime") ~ "Past Crime",
  Final.Incident.Type %in% c("Vehicle Accident") ~
    "Vehicle Accident",
  TRUE ~ "Other",
  # If Incident.Type is NA, assign "Other"
)
)

```

Exploratory Data Analysis

```

# Reproducibility
set.seed(1)

# Make Train and Test using 80/20 split
sample <- sample(c(TRUE, FALSE), nrow(responder_clean),
  replace=TRUE, prob=c(0.8, 0.2))
train <- responder_clean[sample, ]

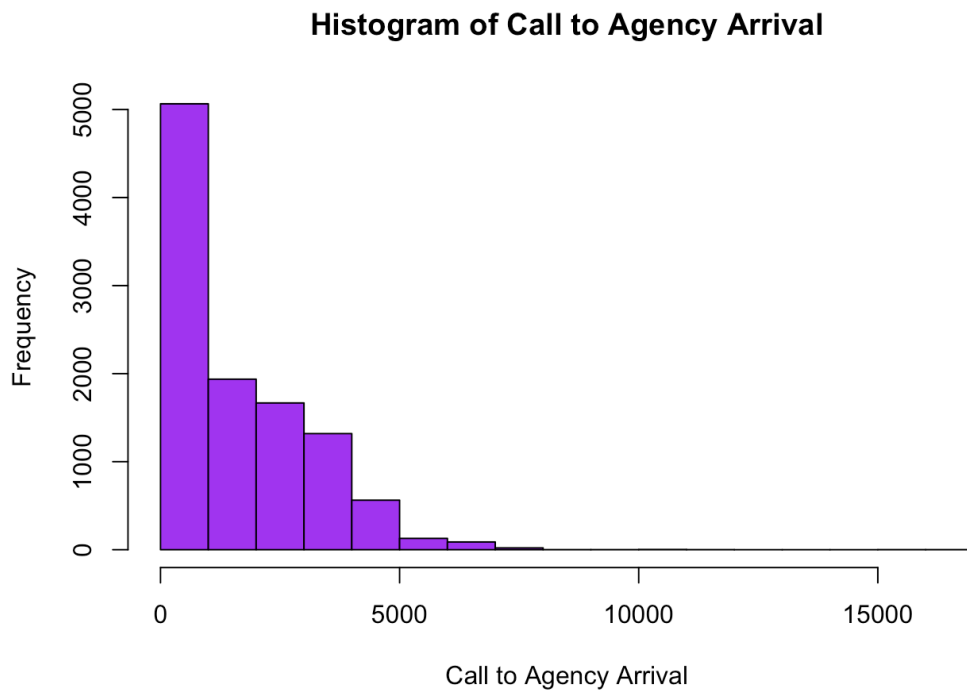
```



```
test <- responder_clean[!sample, ]

# Duplicate train data for further pre-processing
explore<- train

# Create histogram for target variable
hist(Call.to.Agency.Arrival,
      main = "Histogram of Call to Agency Arrival",
      xlab = "Call to Agency Arrival",
      col = "purple",
      border = "black")
```



```
#####
# The target variable is skewed right, indicating#
# the potential for log transformation          #
#####

# Create a new variable that is the log of
# 'Call.to.Agency.Arrival'
explore$Log.Call.to.Agency.Arrival <-
  log(explore$Call.to.Agency.Arrival)

# Check the result by viewing the first few rows
head(explore[, c("Call.to.Agency.Arrival",
                 "Log.Call.to.Agency.Arrival")])
```

Call.to.Agency.Arrival Log.Call.to.Agency.Arrival

1	1719.06	7.449533
2	3254.60	8.087825
3	546.17	6.302930
5	991.80	6.899521
8	3039.50	8.019448
9	666.44	6.501950

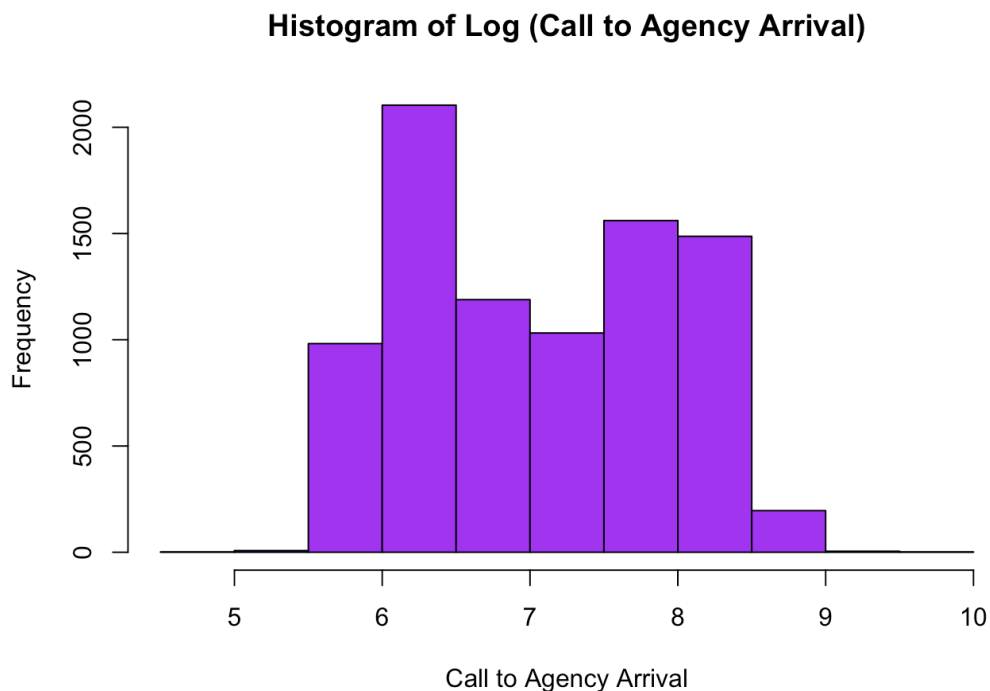
```
# Check for NA values in the log-transformed column
sum(is.na(explore$Log.Call.to.Agency.Arrival))
```

```
[1] 0
```

```
# Check for NA values in the entire data frame
sum(is.na(explore))
```

```
[1] 52
```

```
# Histogram for log transformed target variable
hist(explore$Log.Call.to.Agency.Arrival,
      main = "Histogram of Log (Call to Agency Arrival)",
      xlab = "Call to Agency Arrival",
      col = "purple",
      border = "black")
```



```
# correlation matrix with log target variable
```

```
cor_matrix_log <- cor(explore[,apply(explore,is.numeric)])

# install.packages("corrplot")
library(corrplot)
```

corrplot 0.95 loaded

```
# Save the plot with larger dimensions
png("corrplot_log.png", width = 4000, height = 3200)
corrplot(cor_matrix_log, method = "color", type = "upper",
          addCoef.col = "black",
          number.cex = 4,
          tl.cex = 4)
# Close the plotting device
dev.off()
```

quartz_off_screen

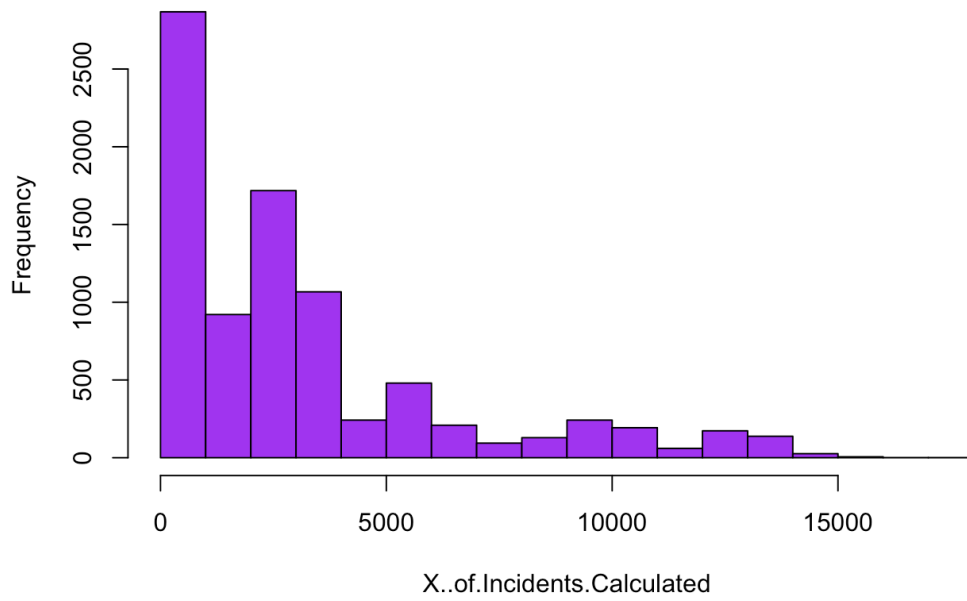
2

```
# Define the dataset variables (excluding the first three)
variables <- c(
  "X..of.Incidents.Calculated",
  "Call.to.First.Pickup",
  "Call.to.PD.Calltaker.Handoff",
  "Call.to.Agency.Dispatch",
  "Call.to.Agency.Arrival",
  "Call.to.First.Arrival..Multi.Agency.Incidents.",
  "Median.Pickup",
  "Median.Calltaker.Handoff",
  "Median.Dispatch",
  "Median.Travel",
  "Median.Cumulative.First.Arrival...Multi.Agency.Incidents.",
  "Average.Travel",
  "Average.Dispatch",
  "Average.Calltaker.Processing",
  "Average.Pickup",
  "Log.Call.to.Agency.Arrival"
)

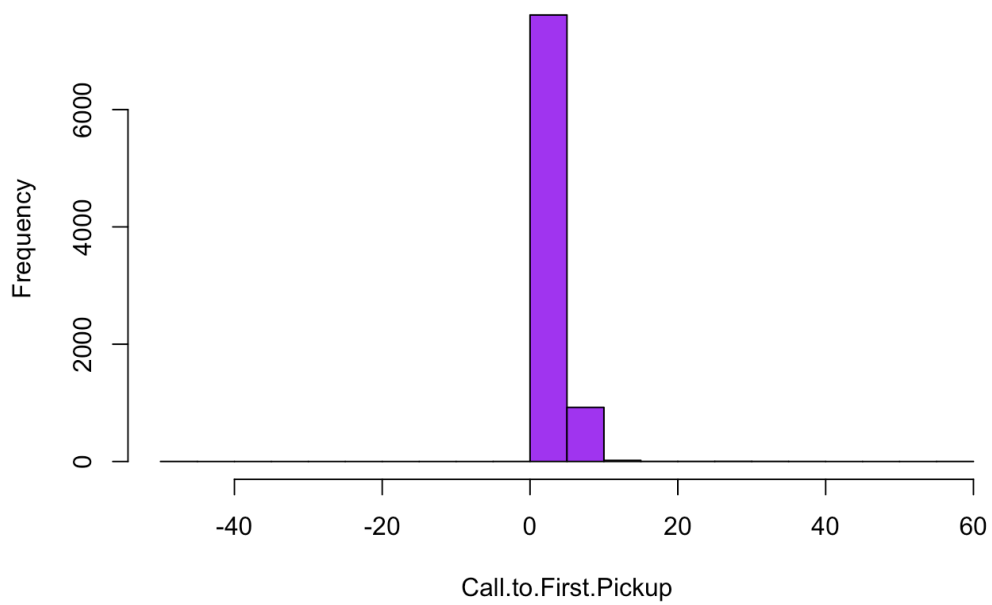
# Loop through the variables and plot histograms for each
variable
for (var in variables) {
  hist(explore[[var]],
        main = paste("Histogram of", var),
        xlab = var,
        col = "purple",
```

```
}  
border = "black")
```

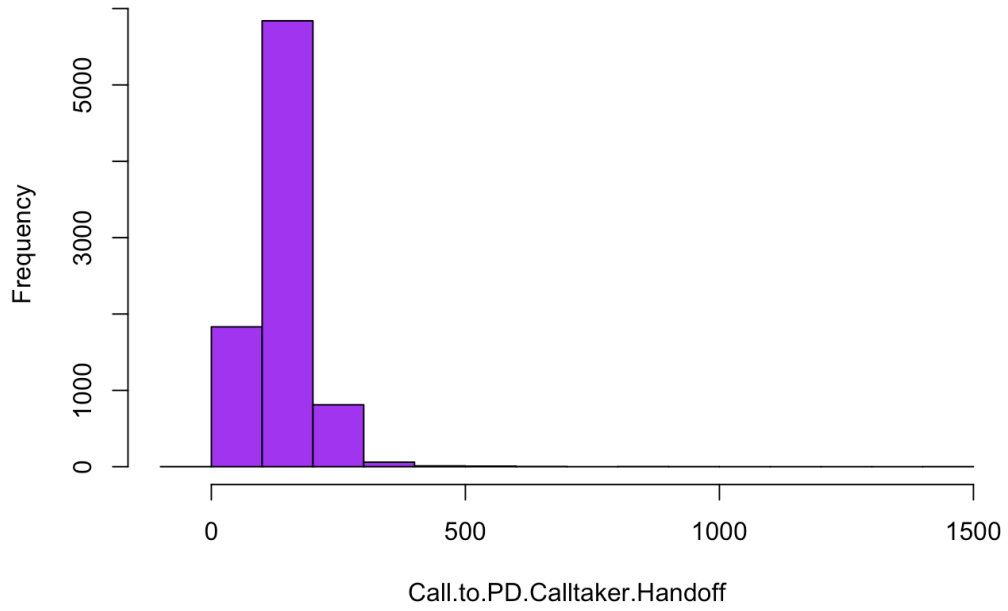
Histogram of X..of.Incidents.Calculated



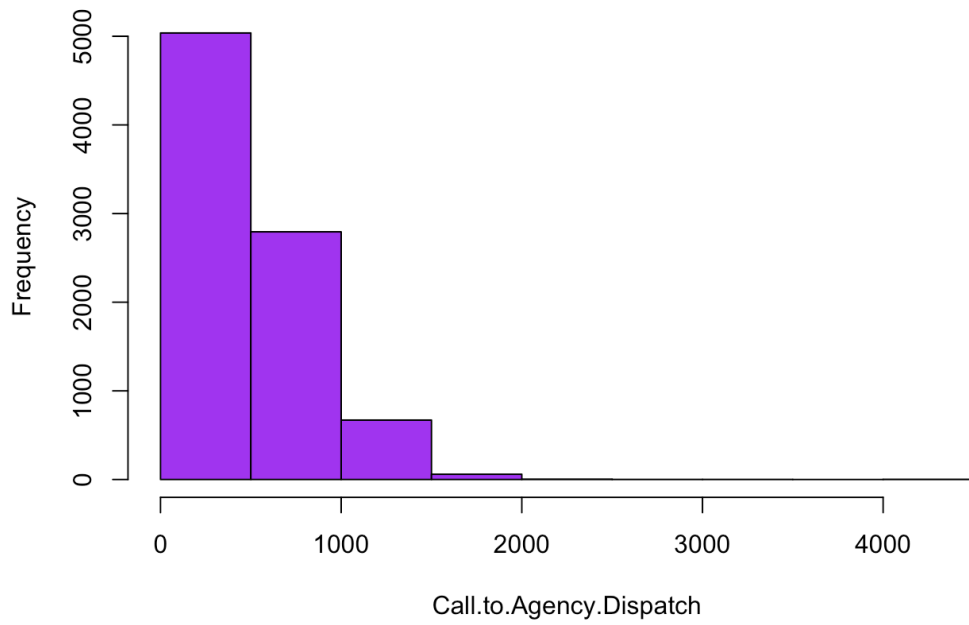
Histogram of Call.to.First.Pickup



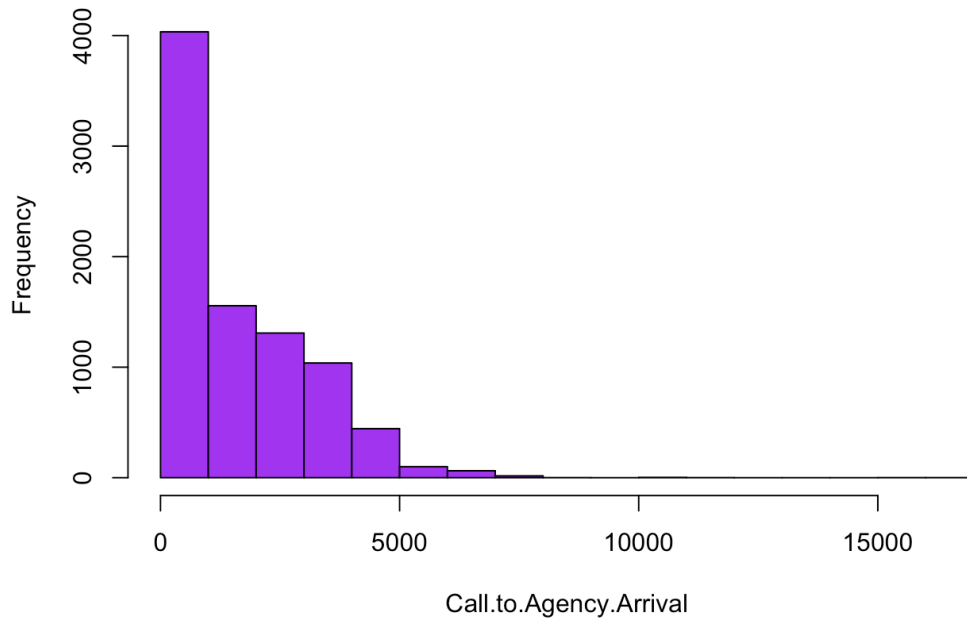
Histogram of Call.to.PD.Calltaker.Handoff



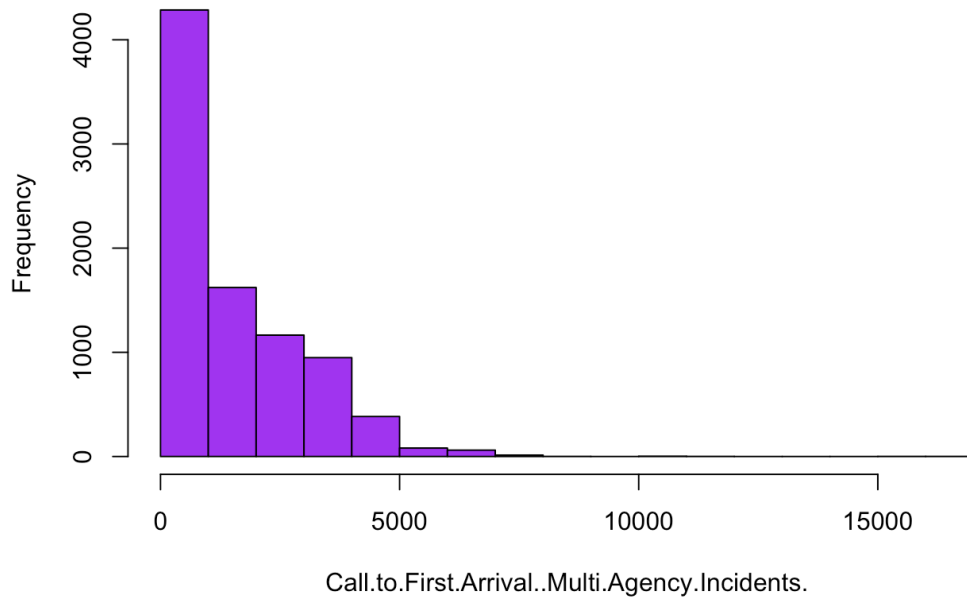
Histogram of Call.to.Agency.Dispatch



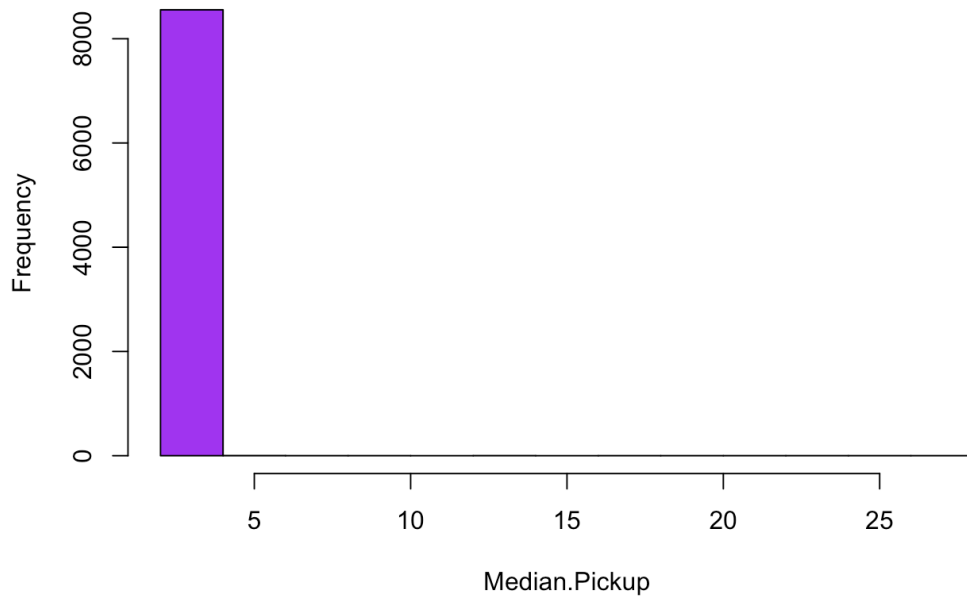
Histogram of Call.to.Agency.Arrival



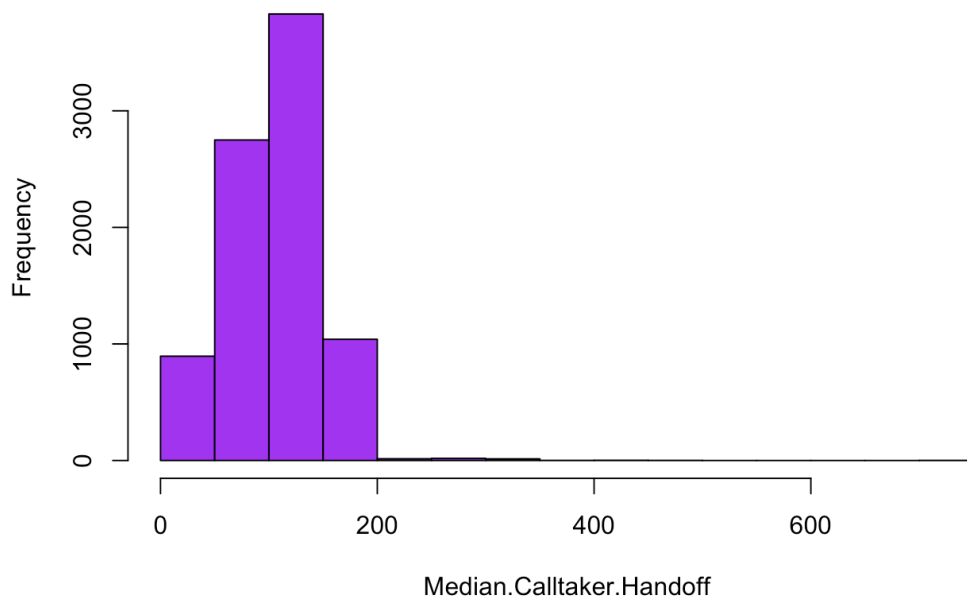
Histogram of Call.to.First.Arrival..Multi.Agency.Incidents.



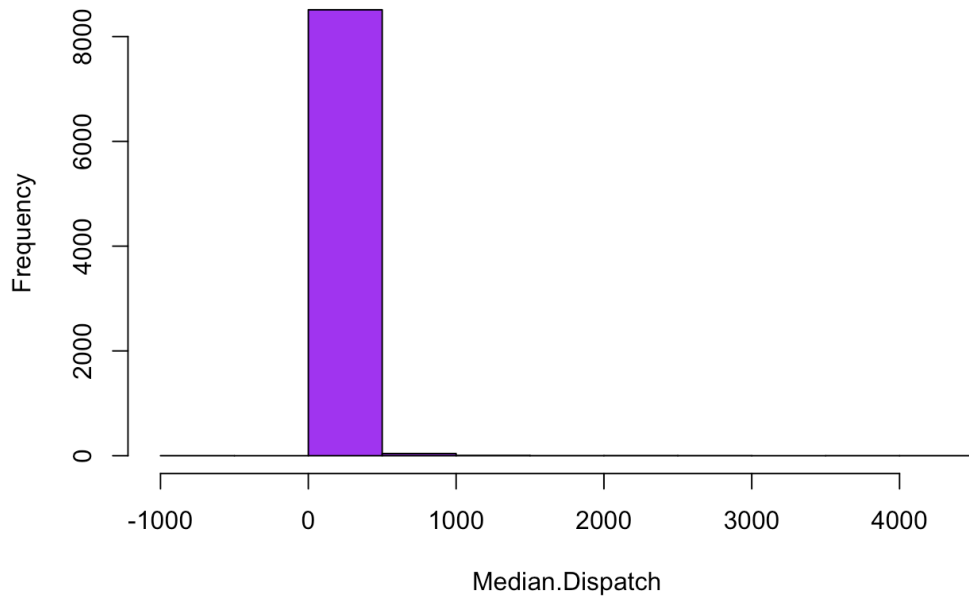
Histogram of Median.Pickup



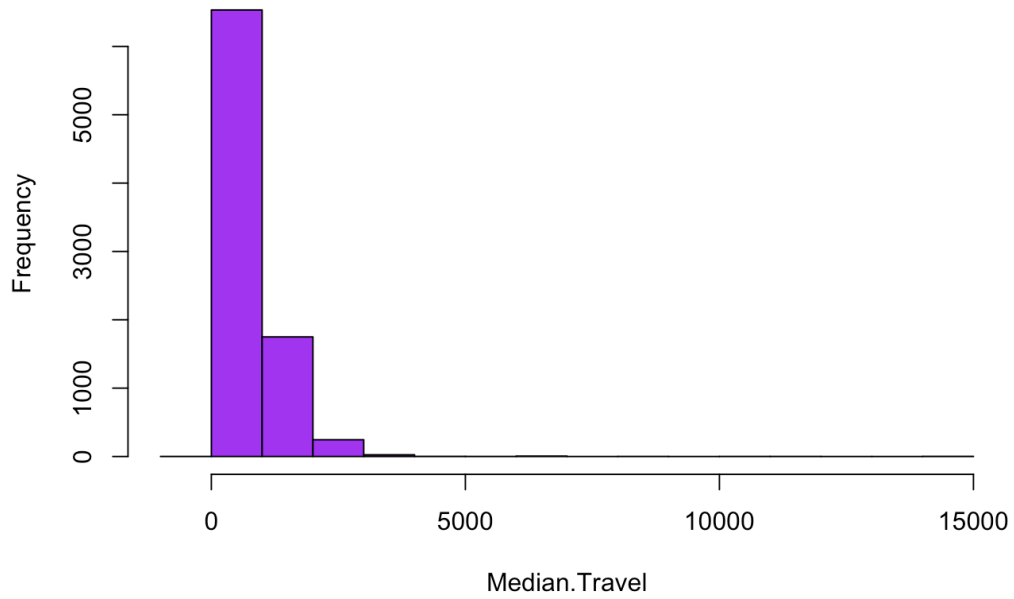
Histogram of Median.Calltaker.Handoff



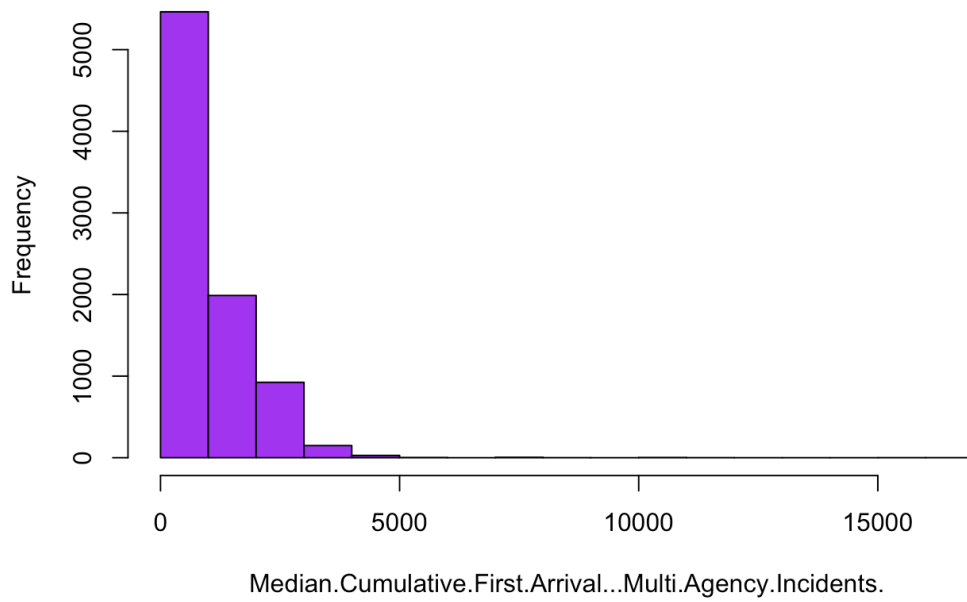
Histogram of Median.Dispatch



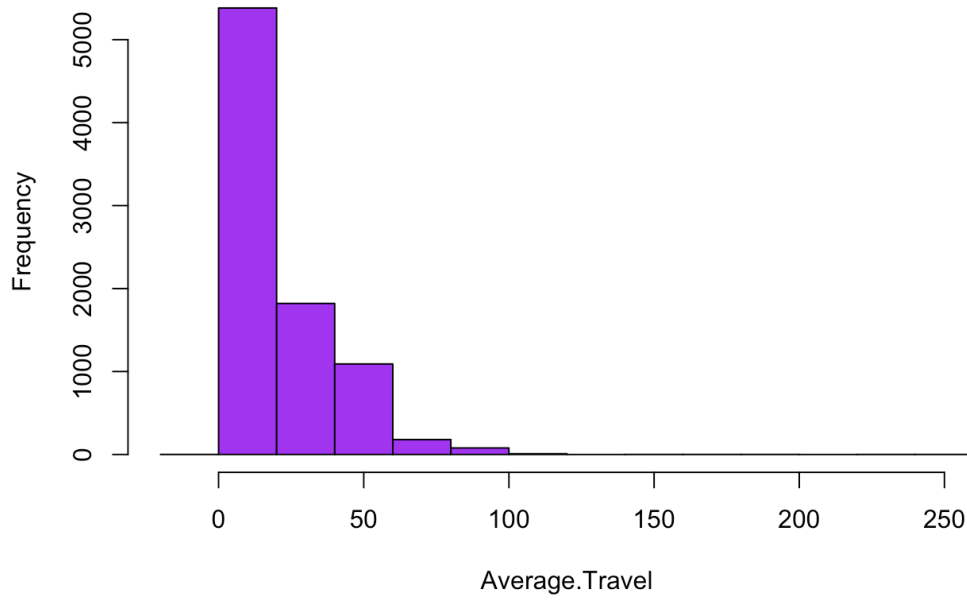
Histogram of Median.Travel



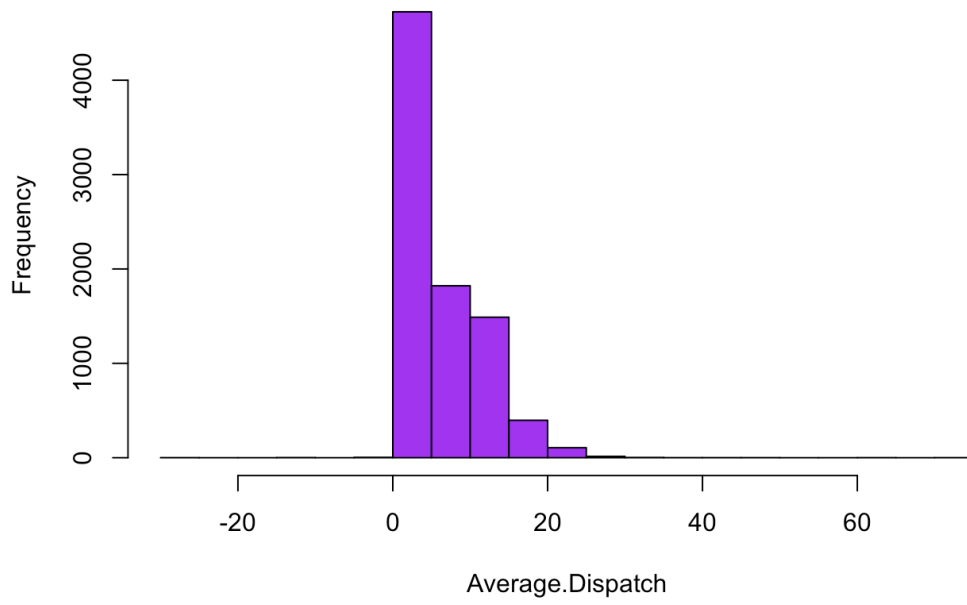
Histogram of Median.Cumulative.First.Arrival...Multi.Agency.Incidents



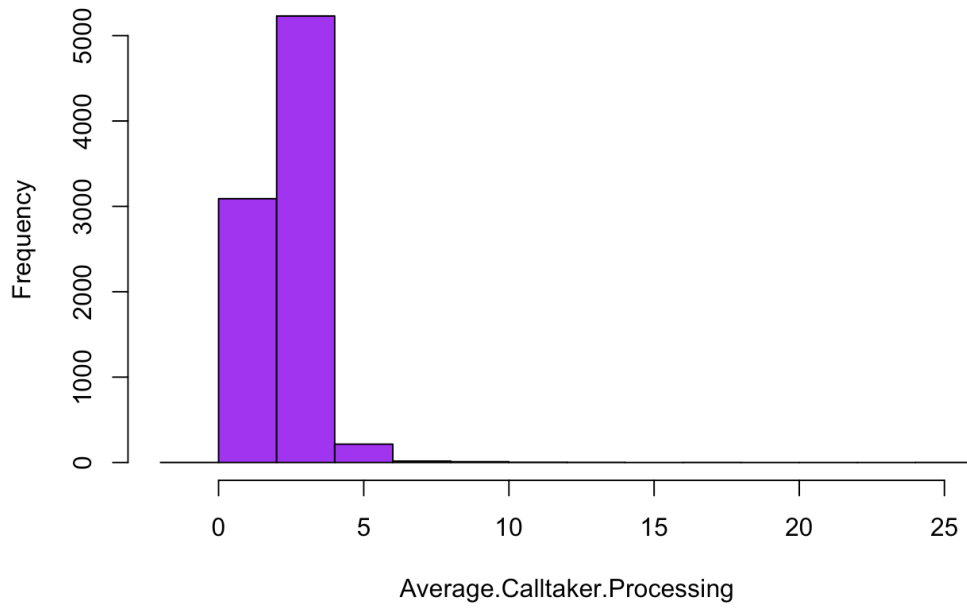
Histogram of Average.Travel



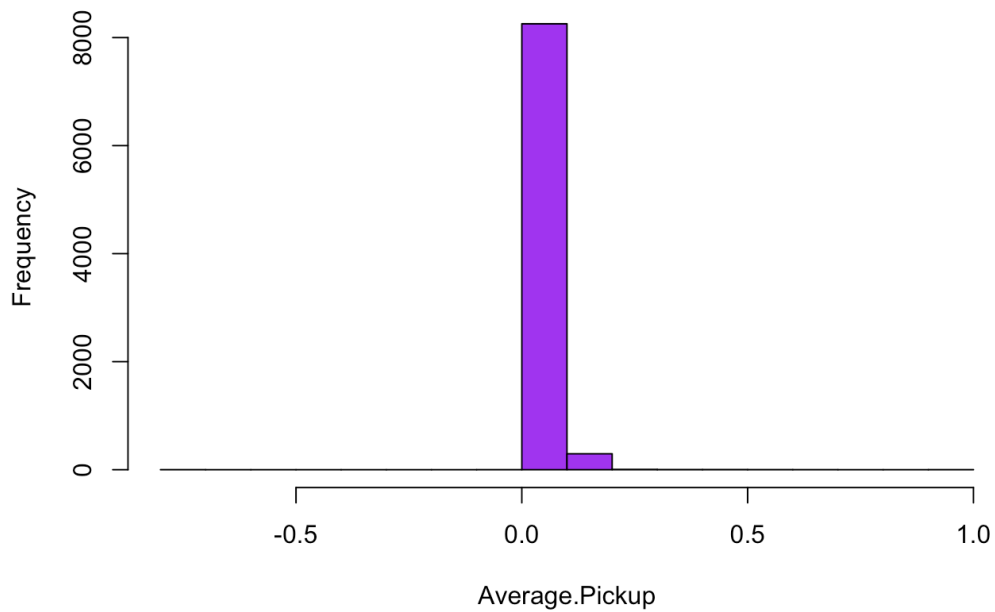
Histogram of Average.Dispatch



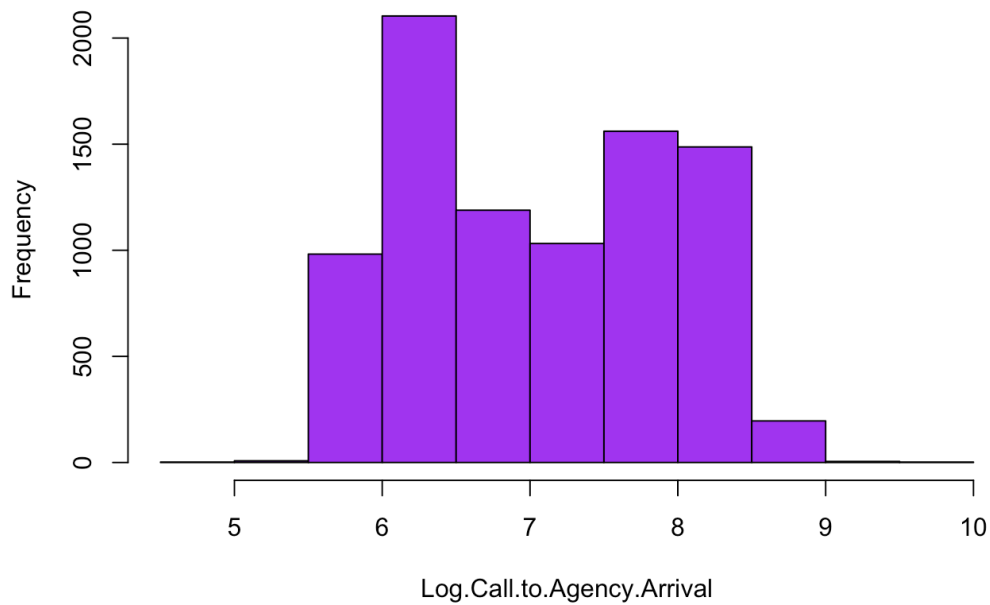
Histogram of Average.Calltaker.Processing



Histogram of Average.Pickup



Histogram of Log.Call.to.Agency.Arrival



```
#####  
# Most variables are skewed right, indicating#  
# the potential for log transformation      #  
#####  
  
# Define the dataset variables  
log_variables <- c(
```

```

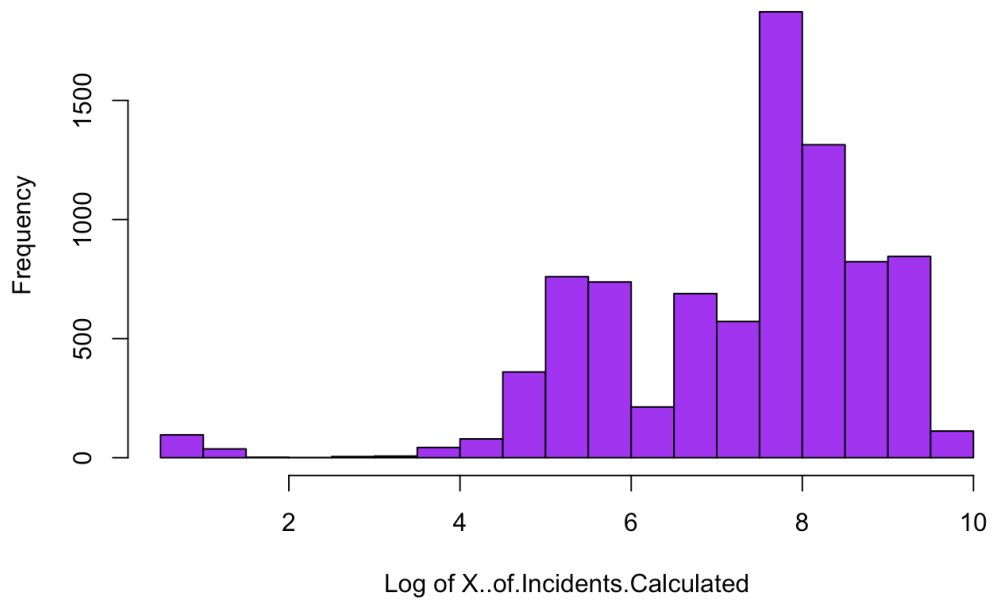
"X..of.Incidents.Calculated",
"Call.to.First.Pickup",
"Call.to.PD.Calltaker.Handoff",
"Call.to.Agency.Dispatch",
"Call.to.Agency.Arrival",
"Call.to.First.Arrival..Multi.Agency.Incidents.",
"Median.Pickup",
"Median.Calltaker.Handoff",
"Median.Dispatch",
"Median.Travel",
"Median.Cumulative.First.Arrival...Multi.Agency.Incidents.",
"Average.Travel",
"Average.Dispatch",
"Average.Calltaker.Processing",
"Average.Pickup"
)

# Loop through the variables, take the log, and plot
  histograms
# Adding 1 to avoid log(0)
for (var in log_variables) {
  log_var <- log(explore[[var]] + 1)

  # Create the histogram for the log-transformed variable
  hist(log_var,
        main = paste("Histogram of Log", var),
        xlab = paste("Log of", var),
        col = "purple",
        border = "black")
}

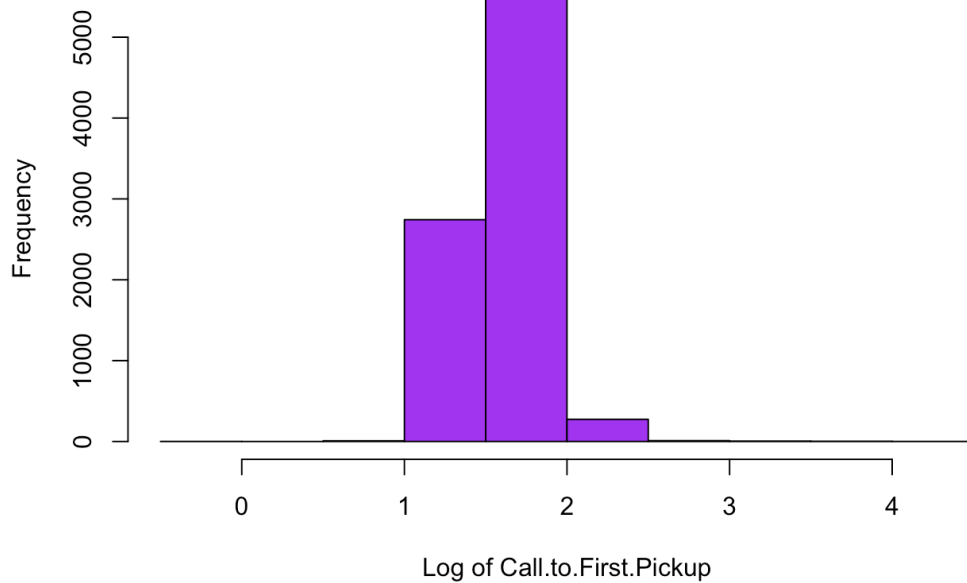
```

Histogram of Log X..of.Incidents.Calculated

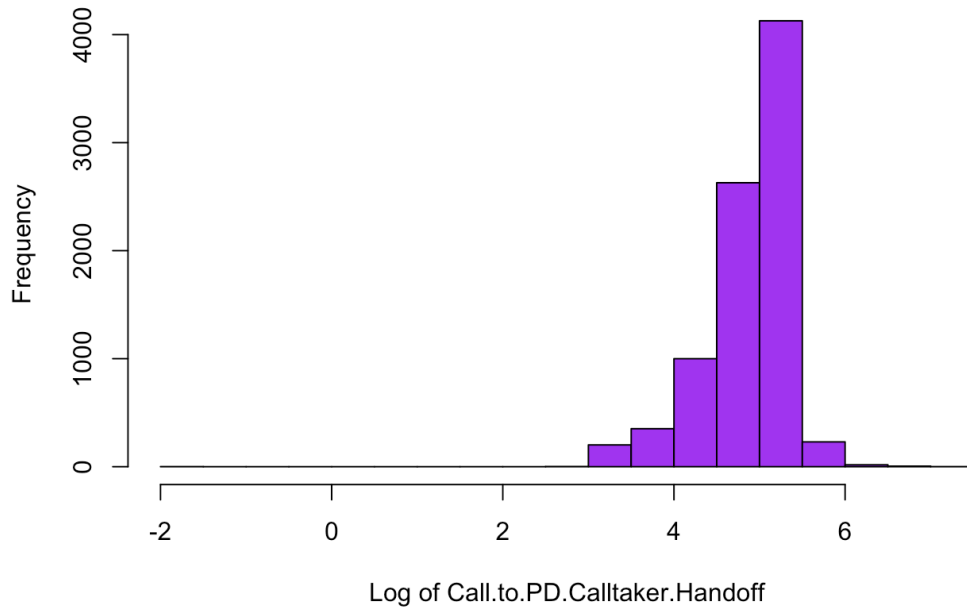


Warning in log(explore[[var]] + 1): NaNs produced

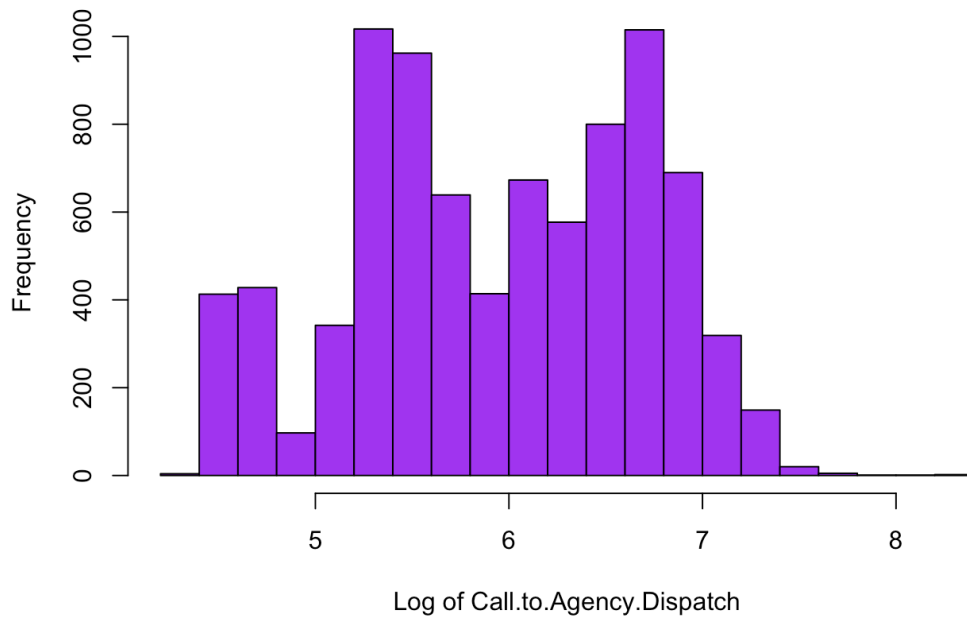
Histogram of Log Call.to.First.Pickup



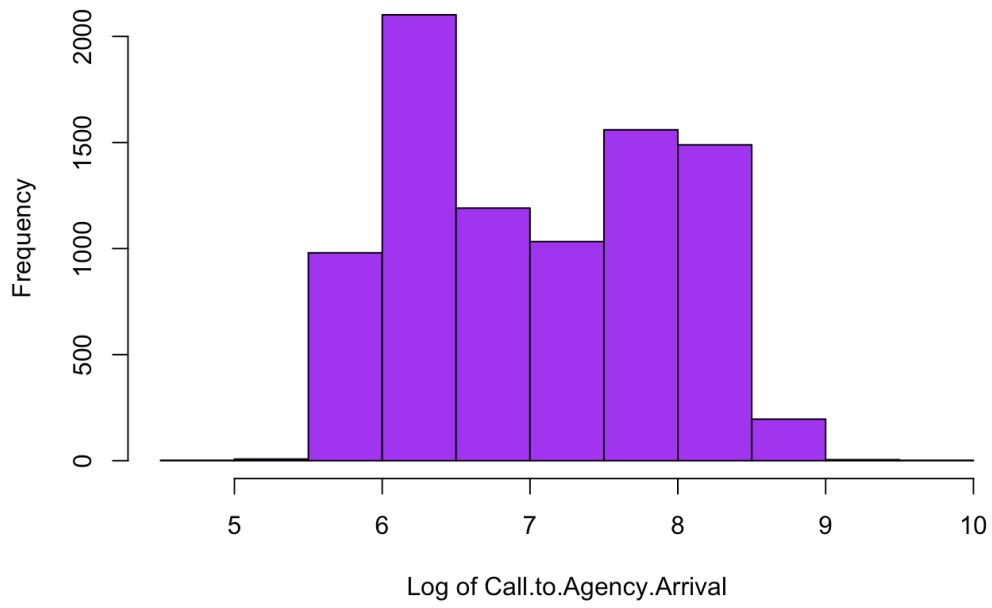
Histogram of Log Call.to.PD.Calltaker.Handoff



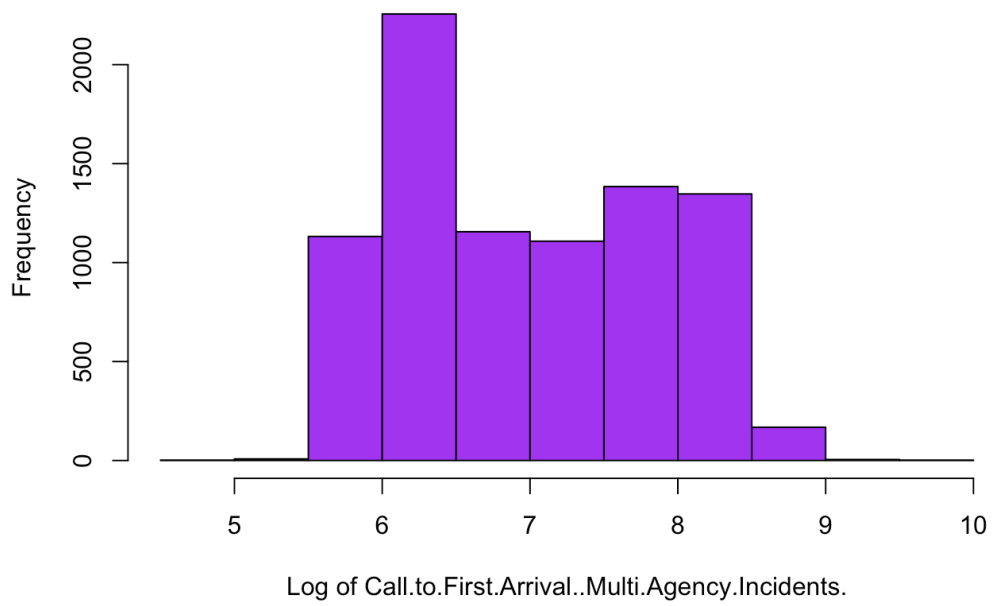
Histogram of Log Call.to.Agency.Dispatch



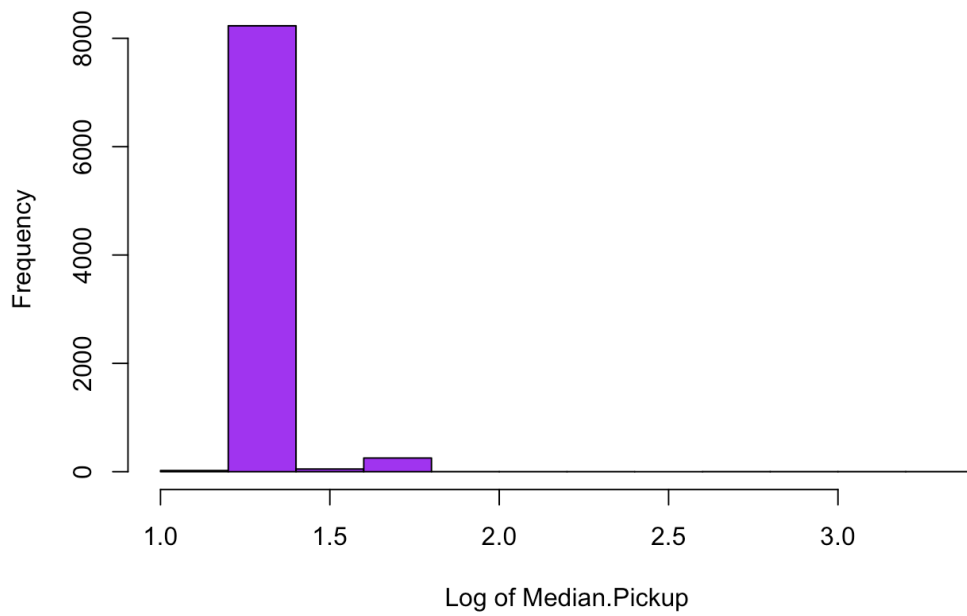
Histogram of Log Call.to.Agency.Arrival



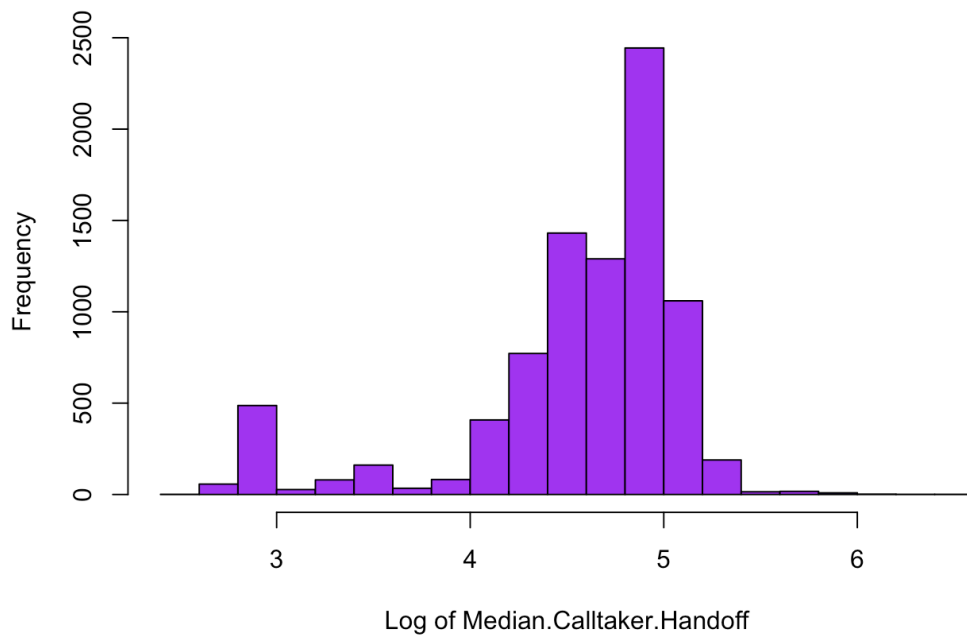
Histogram of Log Call.to.First.Arrival..Multi.Agency.Incidents.



Histogram of Log Median.Pickup

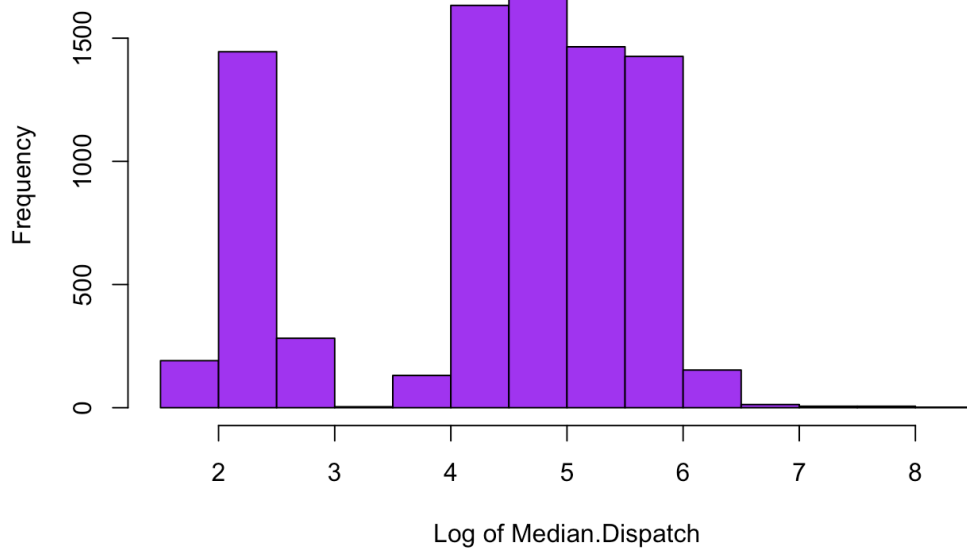


Histogram of Log Median.Calltaker.Handoff



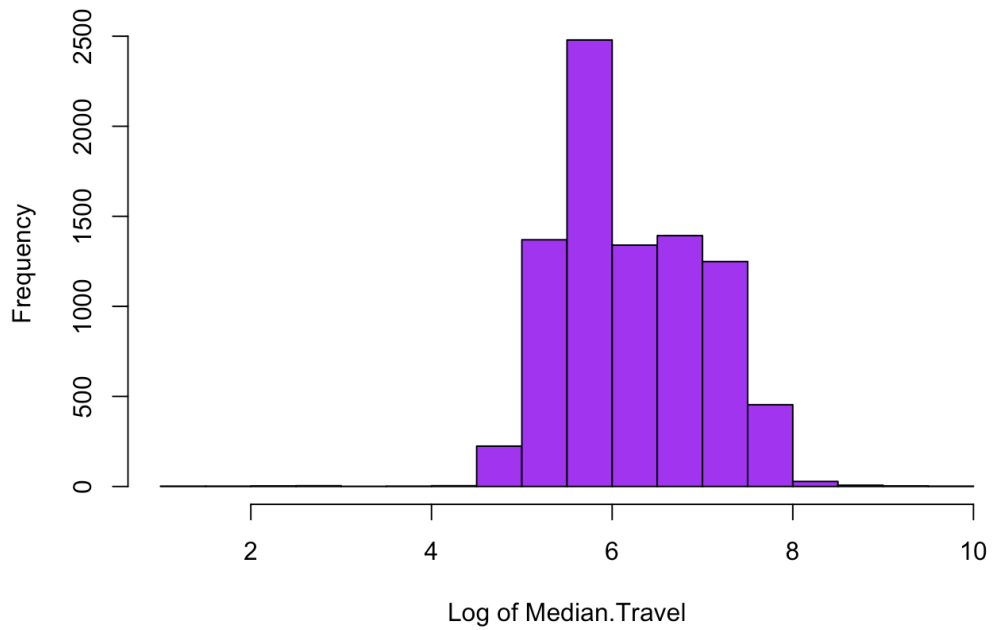
Warning in log(explore[[var]] + 1): NaNs produced

Histogram of Log Median.Dispatch

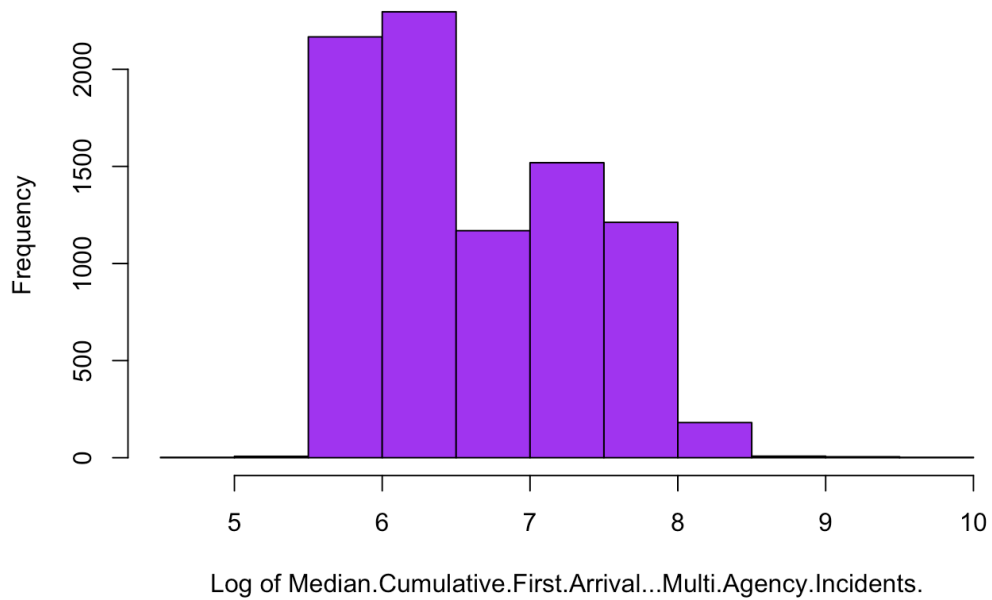


Warning in log(explore[[var]] + 1): NaNs produced

Histogram of Log Median.Travel

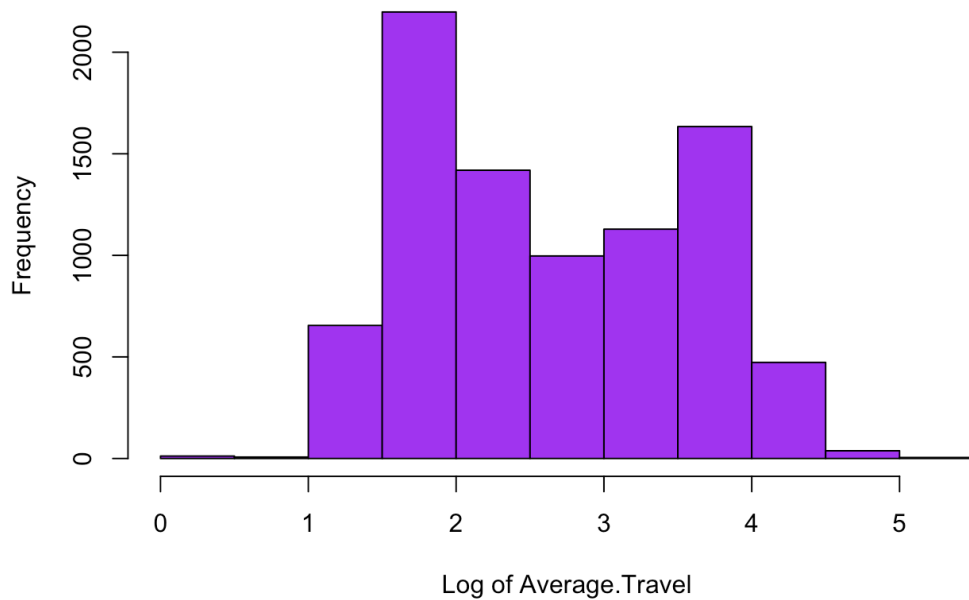


Histogram of Log Median.Cumulative.First.Arrival...Multi.Agency.Inciden



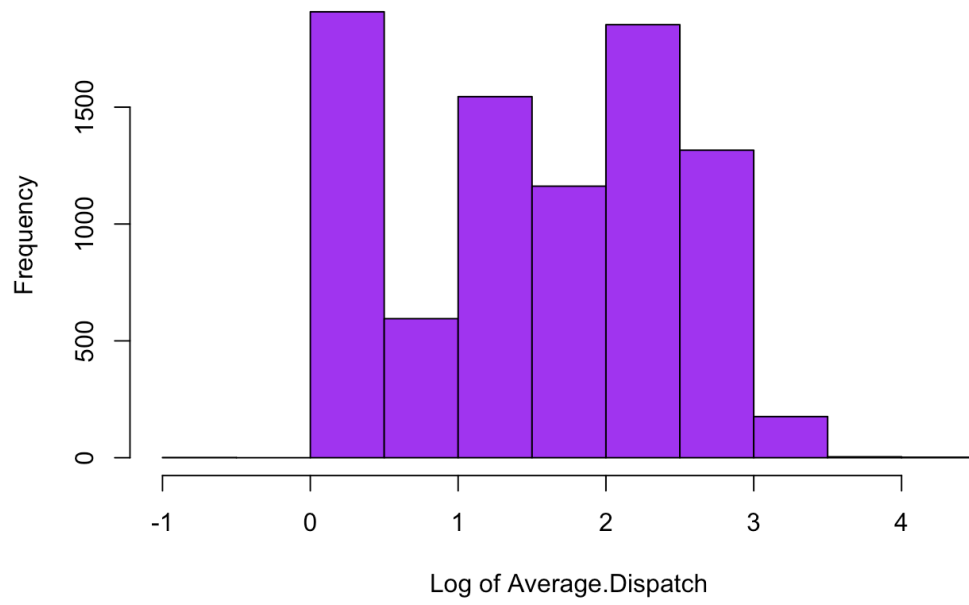
Warning in log(explore[[var]] + 1): NaNs produced

Histogram of Log Average.Travel

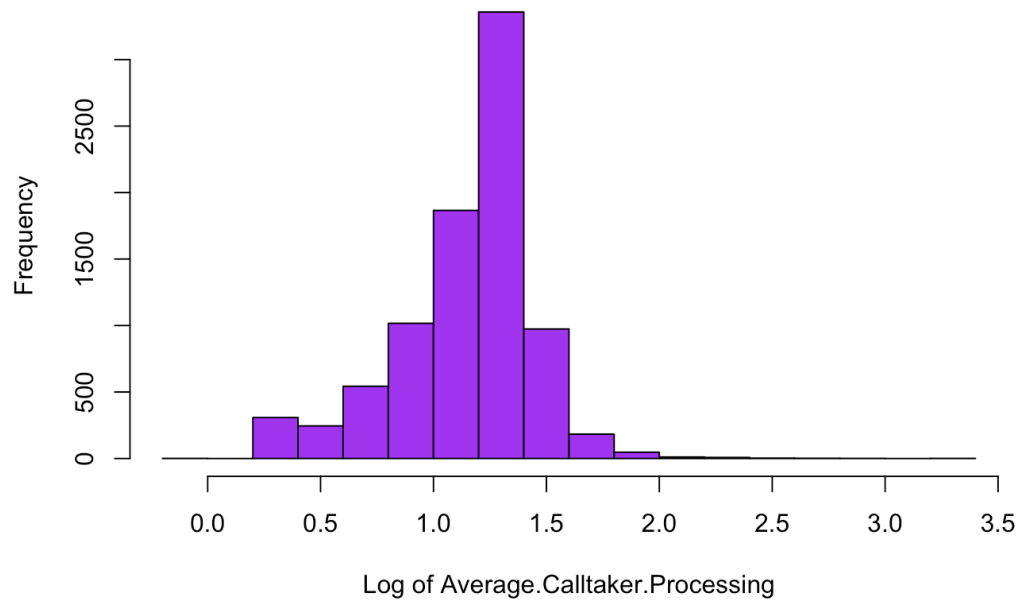


Warning in log(explore[[var]] + 1): NaNs produced

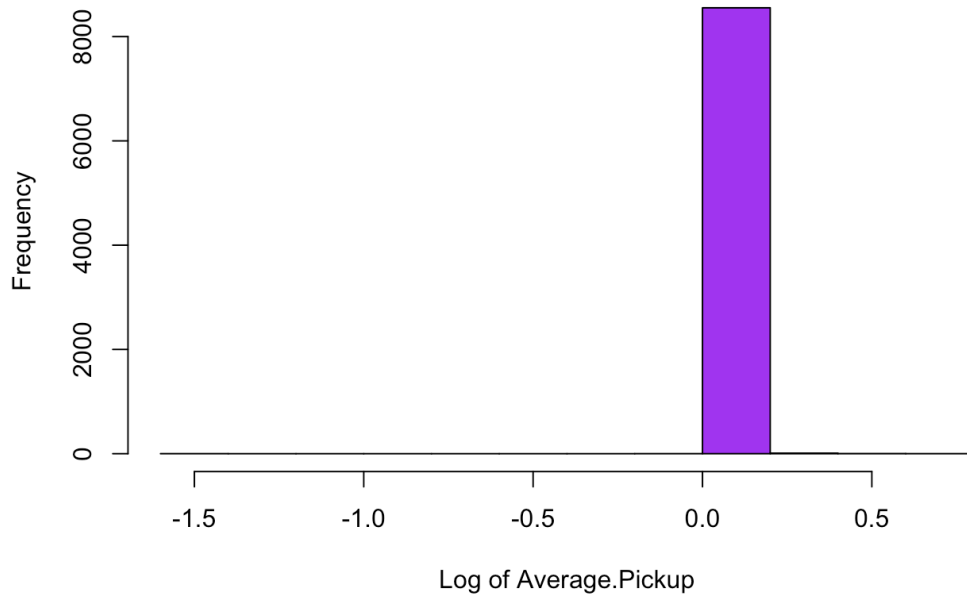
Histogram of Log Average.Dispatch



Histogram of Log Average.Calltaker.Processing

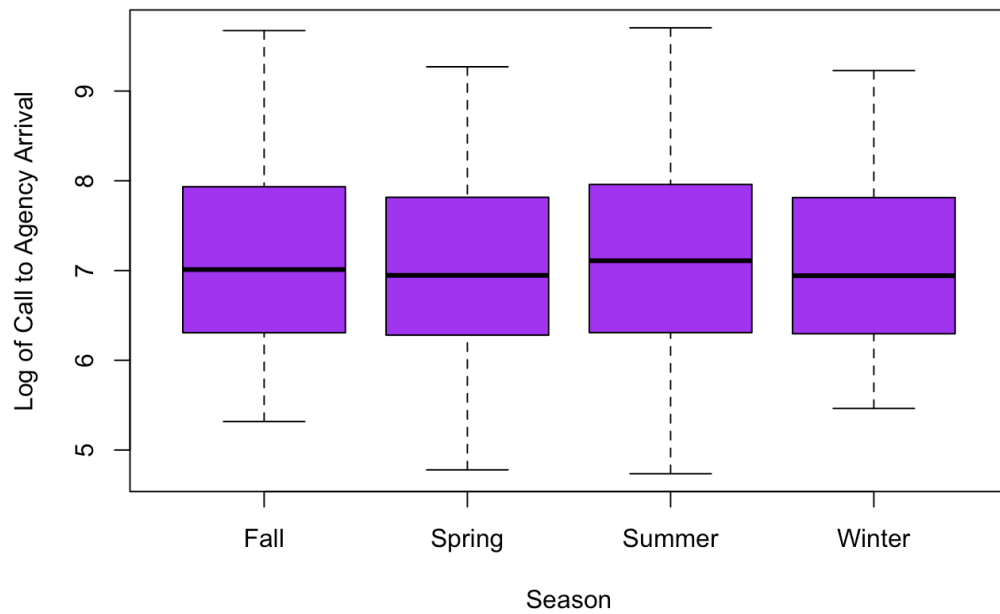


Histogram of Log Average.Pickup



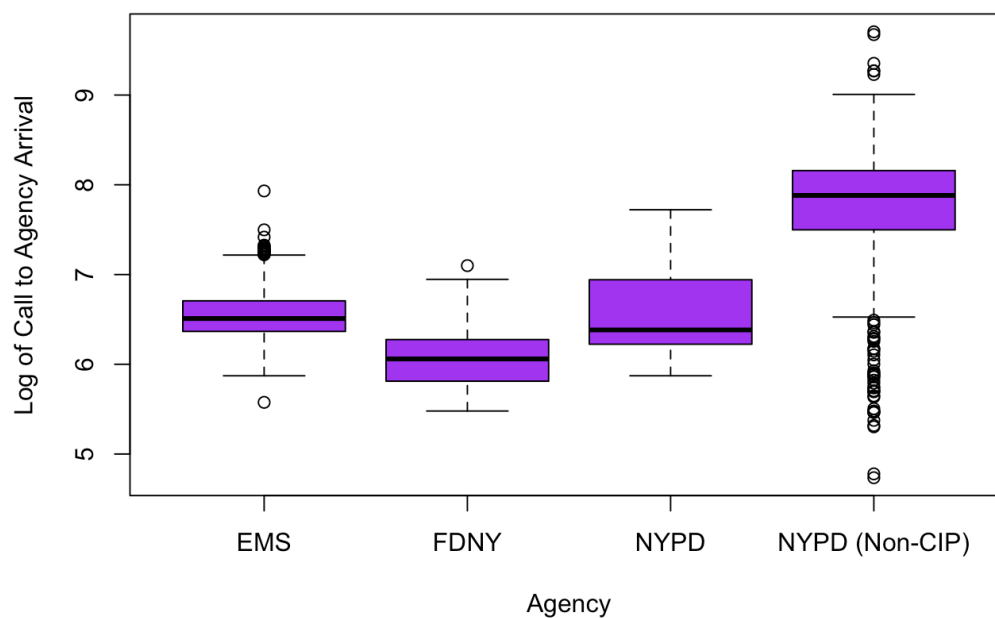
```
# Side-by-side boxplot of the log-transformed variable by
  Season
boxplot(Log.Call.to.Agency.Arrival ~ Season,
  data = explore,
  main = "Side-by-Side Boxplot by Season",
  xlab = "Season",
  ylab = "Log of Call to Agency Arrival",
  col = "purple",
  border = "black")
```

Side-by-Side Boxplot by Season



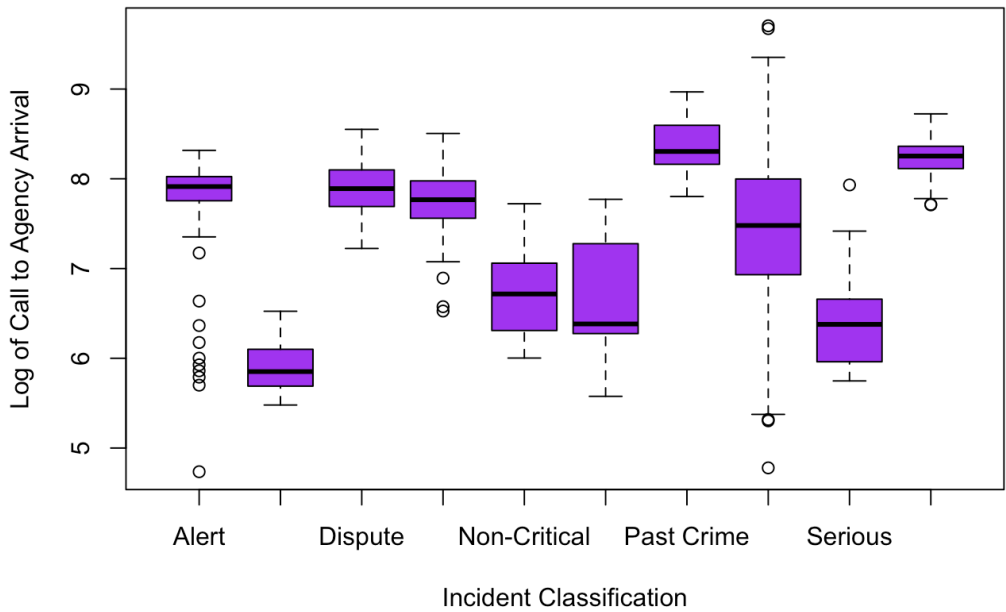
```
# Side-by-side boxplot of the log-transformed variable by
Agency
boxplot(Log.Call.to.Agency.Arrival ~ Agency,
        data = explore,
        main = "Side-by-Side Boxplot by Agency",
        xlab = "Agency",
        ylab = "Log of Call to Agency Arrival",
        col = "purple",
        border = "black")
```

Side-by-Side Boxplot by Agency



```
# Side-by-side boxplot of the log-transformed variable by Incident
# Classification
boxplot(Log.Call.to.Agency.Arrival ~ IncidentClassification,
        data = explore,
        main = "Side-by-Side Boxplot by Incident Classification",
        xlab = "Incident Classification",
        ylab = "Log of Call to Agency Arrival",
        col = "purple",
        border = "black")
```

Side-by-Side Boxplot by Incident Classification



```
# ANOVA for Incident Classification
IC_Anova <- aov(Call.to.Agency.Arrival~IncidentClassification,
               data=train)
summary(IC_Anova)
```

```
              Df    Sum Sq   Mean Sq F value Pr(>F)
IncidentClassification    9 1.237e+10  1.374e+09   2390 <2e-16
***
Residuals              8558  4.922e+09   5.751e+05
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
IC_ttest<- TukeyHSD(IC_Anova)
IC_ttest
```

Tukey multiple comparisons of means
95% family-wise confidence level

```
Fit: aov(formula = Call.to.Agency.Arrival ~
IncidentClassification, data = train)
```

```
$IncidentClassification

              diff      lwr      upr      p adj
Critical-Alert -2281.791088 -2416.40073 -2147.18145 0.0000000
```

Dispute-Alert	106.663662	-28.82897
242.15629 0.2731452		
Hazard-Alert	-169.650353	-324.39757
-14.90314 0.0187637		
Non-Critical-Alert	-1756.228679	-1890.88715
-1621.57021 0.0000000		
Other-Alert	-1762.733792	-1888.79394
-1636.67364 0.0000000		
Past Crime-Alert	1753.045866	1596.62927
1909.46247 0.0000000		
Possible Crime-Alert	-472.187214	-604.82229
-339.55213 0.0000000		
Serious-Alert	-2034.551482	-2161.31700
-1907.78596 0.0000000		
Vehicle Accident-Alert	1185.474807	1029.82479
1341.12483 0.0000000		
Dispute-Critical	2388.454750	2276.51143
2500.39807 0.0000000		
Hazard-Critical	2112.140735	1977.53109
2246.75038 0.0000000		
Non-Critical-Critical	525.562409	414.63019
636.49463 0.0000000		
Other-Critical	519.057296	418.73673
619.37786 0.0000000		
Past Crime-Critical	4034.836954	3898.31147
4171.36244 0.0000000		
Possible Crime-Critical	1809.603874	1701.13674
1918.07101 0.0000000		
Serious-Critical	247.239607	146.03411
348.44510 0.0000000		
Vehicle Accident-Critical	3467.265895	3331.61936
3602.91243 0.0000000		
Hazard-Dispute	-276.314015	-411.80665
-140.82138 0.0000000		
Non-Critical-Dispute	-1862.892341	-1974.89437
-1750.89031 0.0000000		
Other-Dispute	-1869.397454	-1970.89973
-1767.89517 0.0000000		
Past Crime-Dispute	1646.382204	1508.98605
1783.77836 0.0000000		
Possible Crime-Dispute	-578.850876	-688.41189
-469.28986 0.0000000		
Serious-Dispute	-2141.215143	-2243.59214
-2038.83815 0.0000000		
Vehicle Accident-Dispute	1078.811145	942.28833
1215.33396 0.0000000		
Non-Critical-Hazard	-1586.578326	-1721.23680
-1451.91986 0.0000000		

Other-Hazard	-1593.083439	-1719.14359
-1467.02329 0.0000000		
Past Crime-Hazard	1922.696220	1766.27962
2079.11282 0.0000000		
Possible Crime-Hazard	-302.536861	-435.17194
-169.90178 0.0000000		
Serious-Hazard	-1864.901128	-1991.66665
-1738.13561 0.0000000		
Vehicle Accident-Hazard	1355.125161	1199.47514
1510.77518 0.0000000		
Other-Non-Critical	-6.505113	-106.89118
93.88096 1.0000000		
Past Crime-Non-Critical	3509.274546	3372.70092
3645.84817 0.0000000		
Possible Crime-Non-Critical	1284.041465	1175.51374
1392.56919 0.0000000		
Serious-Non-Critical	-278.322802	-379.59323
-177.05238 0.0000000		
Vehicle Accident-Non-Critical	2941.703487	2806.00849
3077.39848 0.0000000		
Past Crime-Other	3515.779658	3387.67574
3643.88357 0.0000000		
Possible Crime-Other	1290.546578	1192.89145
1388.20171 0.0000000		
Serious-Other	-271.817689	-361.33840
-182.29698 0.0000000		
Vehicle Accident-Other	2948.208600	2821.04182
3075.37538 0.0000000		
Possible Crime-Past Crime	-2225.233080	-2359.81211
-2090.65405 0.0000000		
Serious-Past Crime	-3787.597348	-3916.39544
-3658.79926 0.0000000		
Vehicle Accident-Past Crime	-567.571059	-724.88088
-410.26124 0.0000000		
Serious-Possible Crime	-1562.364267	-1660.92826
-1463.80028 0.0000000		
Vehicle Accident-Possible Crime	1657.662021	1523.97473
1791.34931 0.0000000		
Vehicle Accident-Serious	3220.026289	3092.16025
3347.89233 0.0000000		

```
# ANOVA for Agency
```

```
Agency_Anova <- aov(Call.to.Agency.Arrival~Agency, data=train)
summary(Agency_Anova)
```

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
Agency	3	9.923e+09	3.308e+09	3844	<2e-16 ***

```
Residuals    8564 7.369e+09 8.604e+05
```

```
----
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Agency_ttest<- TukeyHSD(Agency_Anova)
Agency_ttest
```

Tukey multiple comparisons of means
95% family-wise confidence level

```
Fit: aov(formula = Call.to.Agency.Arrival ~ Agency, data =
train)
```

```
$Agency
```

	diff	lwr	upr	p adj
FDNY-EMS	-286.39531	-380.66843	-192.1222	0.0000000
NYPD-EMS	51.73023	-48.20959	151.6701	0.5437247
NYPD (Non-CIP)-EMS	2028.91954	1943.74532	2114.0938	0.0000000
NYPD-FDNY	338.12555	254.36873	421.8824	0.0000000
NYPD (Non-CIP)-FDNY	2315.31486	2249.87874	2380.7510	0.0000000
NYPD (Non-CIP)-NYPD	1977.18931	1903.82424	2050.5544	0.0000000

```
# List of variables to be log-transformed
log_variables <- c(
  "X..of.Incidents.Calculated",
  "Call.to.First.Pickup",
  "Call.to.PD.Calltaker.Handoff",
  "Call.to.Agency.Dispatch",
  "Call.to.First.Arrival..Multi.Agency.Incidents.",
  "Median.Pickup",
  "Median.Calltaker.Handoff",
  "Median.Dispatch",
  "Median.Travel",
  "Median.Cumulative.First.Arrival...Multi.Agency.Incidents.",
  "Average.Travel",
  "Average.Dispatch",
  "Average.Calltaker.Processing",
  "Average.Pickup"
)

# Subset the data to include only numeric variables
log_data <- explore[, log_variables]

# Replace zero or negative values with 0.0001
log_data[log_data <= 0] <- 0.0001
```

```

# Log-transform the selected variables, adding 1 to avoid
  log(0)
log_data_transformed <- log(log_data + 1)

# Add the already log-transformed 'Log.Call.to.Agency.Arrival'
log_data_transformed$Log.Call.to.Agency.Arrival <-
  explore$Log.Call.to.Agency.Arrival

# Numeric columns only
log_data_numeric <-
  log_data_transformed[, sapply(log_data_transformed,
    is.numeric)]

# Create the correlation matrix
cor_matrix_log <- cor(log_data_numeric, use = "complete.obs")

# Save the plot with larger dimensions
png("corrplot_large_log.png", width = 4000, height = 3200)

# Generate the correlation plot
corrplot(cor_matrix_log, method = "color", type = "upper",
  addCoef.col = "black",
  number.cex = 4,
  tl.cex = 4)

dev.off()

```

quartz_off_screen

2

```

#####
# Variables seem to be influenced by extreme #
# outliers; attempting winsorization          #
#####

# install.packages("ggplot2")
# install.packages("reshape2")
library(ggplot2)
library(reshape2)

# List of variables to winsorize
variables <- c( "X..of.Incidents.Calculated",
  "Call.to.First.Pickup",
  "Call.to.PD.Calltaker.Handoff",
  "Log.Call.to.Agency.Arrival",
  "Call.to.Agency.Dispatch",

  "Call.to.First.Arrival..Multi.Agency.Incidents.",

```

```

"Median.Pickup",
"Median.Calltaker.Handoff",
"Median.Dispatch",
"Median.Travel",

"Median.Cumulative.First.Arrival...Multi.Agency.Inciden
"Average.Travel",
"Average.Dispatch",
"Average.Calltaker.Processing",
"Average.Pickup")

# Plot boxplots for all variables
explore_long <- log_data_transformed[variables]

# Reshape data to long format for ggplot
explore_long <- melt(explore_long)

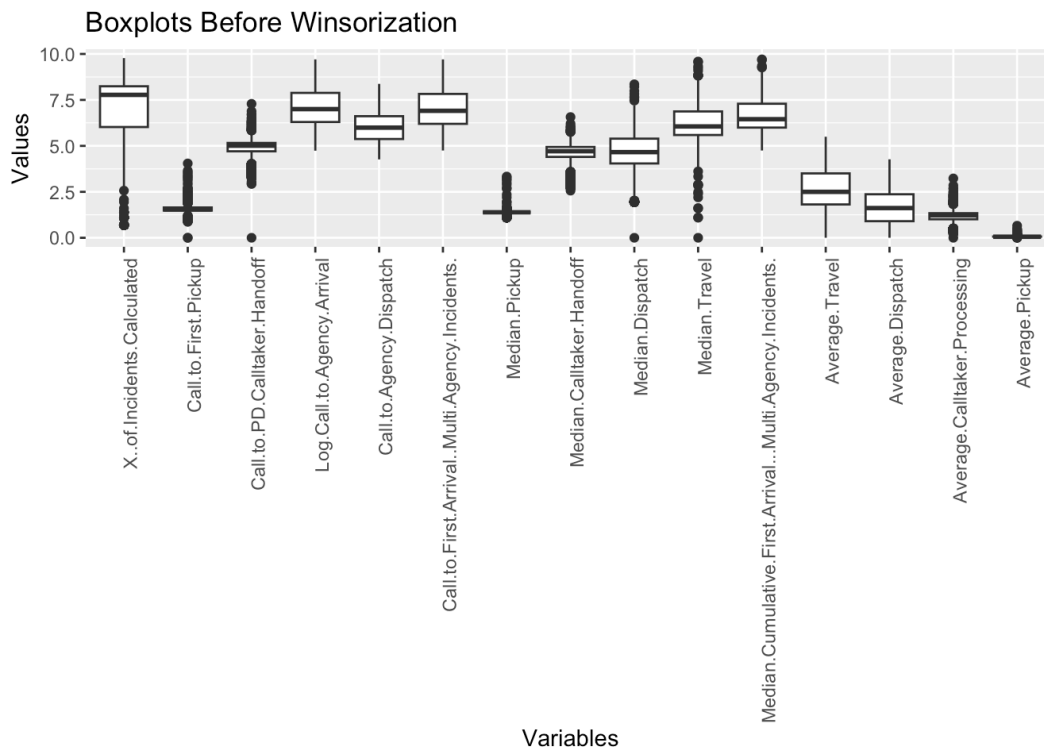
```

No id variables; using all as measure variables

```

ggplot(explore_long, aes(x = variable, y = value)) +
  geom_boxplot() +
  theme(axis.text.x = element_text(angle = 90, hjust = 1)) +
  labs(title = "Boxplots Before Winsorization", x =
        "Variables", y = "Values")

```



```

# Winsorization function

```

```

winsorize <- function(x, lower_percent = 0.01, upper_percent =
  0.99) {
  lower_limit <- quantile(x, lower_percent)
  upper_limit <- quantile(x, upper_percent)
  x[x < lower_limit] <- lower_limit
  x[x > upper_limit] <- upper_limit
  return(x)
}

# Apply winsorization to all variables
log_data_transformed[variables] <-
  lapply(log_data_transformed[variables],
         winsorize)

# Reshape data again for plotting
explore_long_winsorized <- log_data_transformed[variables]
explore_long_winsorized <- melt(explore_long_winsorized)

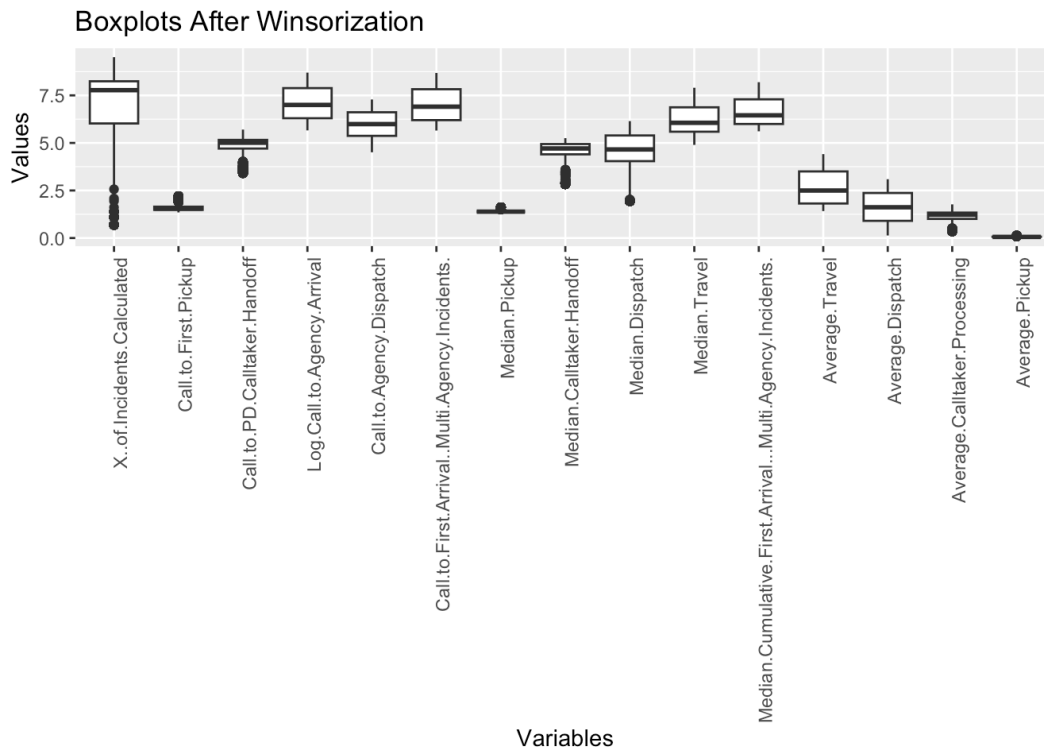
```

No id variables; using all as measure variables

```

# Plot boxplots after winsorization
ggplot(explore_long_winsorized, aes(x = variable, y = value))
  +
  geom_boxplot() +
  theme(axis.text.x = element_text(angle = 90, hjust = 1)) +
  labs(title = "Boxplots After Winsorization", x =
        "Variables", y = "Values")

```



Model Creation

MLR Model

First, a standard MLR was created. Assumptions like normality and homoscedasticity were tested. Unfortunately, this model did not pass the assumptions.

```
# MLR Model
responder_MLR <- lm(
  Log.Call.to.Agency.Arrival ~
    X..of.Incidents.Calculated + Call.to.First.Pickup +
    Call.to.PD.Calltaker.Handoff + Call.to.Agency.Dispatch +
    Call.to.First.Arrival..Multi.Agency.Incidents. +
      Median.Pickup +
    Median.Calltaker.Handoff + Median.Dispatch + Median.Travel
    +
    Median.Cumulative.First.Arrival...Multi.Agency.Incidents. +
    Average.Travel + Average.Dispatch +
      Average.Calltaker.Processing +
    Average.Pickup + as.factor(Agency) +
      as.factor(IncidentClassification),
  data = explore
)

summary(responder_MLR)
```

Call:

```
lm(formula = Log.Call.to.Agency.Arrival ~
  X..of.Incidents.Calculated +
    Call.to.First.Pickup + Call.to.PD.Calltaker.Handoff +
  Call.to.Agency.Dispatch +
    Call.to.First.Arrival..Multi.Agency.Incidents. +
  Median.Pickup +
    Median.Calltaker.Handoff + Median.Dispatch + Median.Travel
  +
    Median.Cumulative.First.Arrival...Multi.Agency.Incidents. +
    Average.Travel + Average.Dispatch +
  Average.Calltaker.Processing +
    Average.Pickup + as.factor(Agency) +
  as.factor(IncidentClassification),
  data = explore)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-1.98511	-0.06187	0.01022	0.07512	0.72449

Coefficients:

Estimate	Std. Error
(Intercept)	
5.833e+00	2.033e-02
X..of.Incidents.Calculated	
3.344e-05	1.086e-06
Call.to.First.Pickup	
-1.086e-02	8.445e-03
Call.to.PD.Calltaker.Handoff	
-2.752e-04	3.524e-04
Call.to.Agency.Dispatch	
2.642e-03	1.128e-04
Call.to.First.Arrival..Multi.Agency.Incidents.	
1.322e-04	2.557e-05
Median.Pickup	
3.282e-05	2.694e-03
Median.Calltaker.Handoff	
-4.377e-04	9.766e-05
Median.Dispatch	
-5.325e-05	2.854e-05
Median.Travel	
-4.353e-05	2.248e-05
Median.Cumulative.First.Arrival...Multi.Agency.Incidents.	
-1.415e-04	2.319e-05
Average.Travel	
1.804e-02	1.432e-03
Average.Dispatch	
-1.257e-01	6.802e-03
Average.Calltaker.Processing	
-5.111e-02	1.989e-02
Average.Pickup	
8.650e-01	5.012e-01
as.factor(Agency)FDNY	
-1.954e-01	1.308e-02
as.factor(Agency)NYPD	
2.211e-01	1.312e-02
as.factor(Agency)NYPD (Non-CIP)	
7.750e-01	1.527e-02
as.factor(IncidentClassification)Critical	
-2.162e-01	1.168e-02
as.factor(IncidentClassification)Dispute	
-9.198e-02	7.845e-03
as.factor(IncidentClassification)Hazard	
1.620e-01	1.901e-02
as.factor(IncidentClassification)Non-Critical	
2.657e-01	1.141e-02

```

as.factor(IncidentClassification)Other
3.340e-02  9.208e-03
as.factor(IncidentClassification)Past Crime
-1.965e-01  1.037e-02
as.factor(IncidentClassification)Possible Crime
-2.469e-01  7.979e-03
as.factor(IncidentClassification)Serious
1.428e-02  1.078e-02
as.factor(IncidentClassification)Vehicle Accident
-3.041e-02  9.807e-03

value Pr(>|t|)
(Intercept)
286.875 < 2e-16 ***
X..of.Incidents.Calculated
30.786 < 2e-16 ***
Call.to.First.Pickup
-1.286 0.198598
Call.to.PD.Calltaker.Handoff
-0.781 0.434929
Call.to.Agency.Dispatch
23.430 < 2e-16 ***
Call.to.First.Arrival..Multi.Agency.Incidents.
5.172 2.37e-07 ***
Median.Pickup
0.012 0.990279
Median.Calltaker.Handoff
-4.482 7.50e-06 ***
Median.Dispatch
-1.866 0.062112 .
Median.Travel
-1.936 0.052876 .
Median.Cumulative.First.Arrival...Multi.Agency.Incidents.
-6.100 1.11e-09 ***
Average.Travel
12.598 < 2e-16 ***
Average.Dispatch
-18.476 < 2e-16 ***
Average.Calltaker.Processing
-2.570 0.010183 *
Average.Pickup
1.726 0.084380 .
as.factor(Agency)FDNY
-14.937 < 2e-16 ***
as.factor(Agency)NYPD
16.858 < 2e-16 ***
as.factor(Agency)NYPD (Non-CIP)
50.753 < 2e-16 ***

```



```

as.factor(IncidentClassification)Critical
-18.505 < 2e-16 ***
as.factor(IncidentClassification)Dispute
-11.724 < 2e-16 ***
as.factor(IncidentClassification)Hazard
8.522 < 2e-16 ***
as.factor(IncidentClassification)Non-Critical
23.296 < 2e-16 ***
as.factor(IncidentClassification)Other
3.627 0.000288 ***
as.factor(IncidentClassification)Past Crime
-18.939 < 2e-16 ***
as.factor(IncidentClassification)Possible Crime
-30.938 < 2e-16 ***
as.factor(IncidentClassification)Serious
1.325 0.185128
as.factor(IncidentClassification)Vehicle Accident
-3.100 0.001939 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

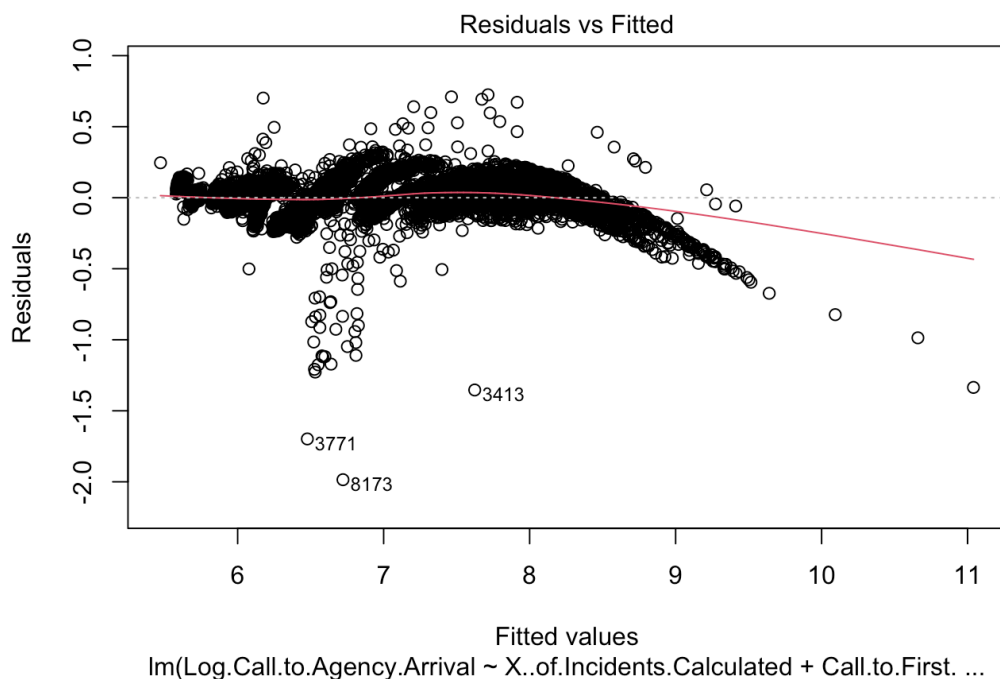
```

Residual standard error: 0.1289 on 8541 degrees of freedom
 Multiple R-squared: 0.9782, Adjusted R-squared: 0.9781
 F-statistic: 1.472e+04 on 26 and 8541 DF, p-value: < 2.2e-16

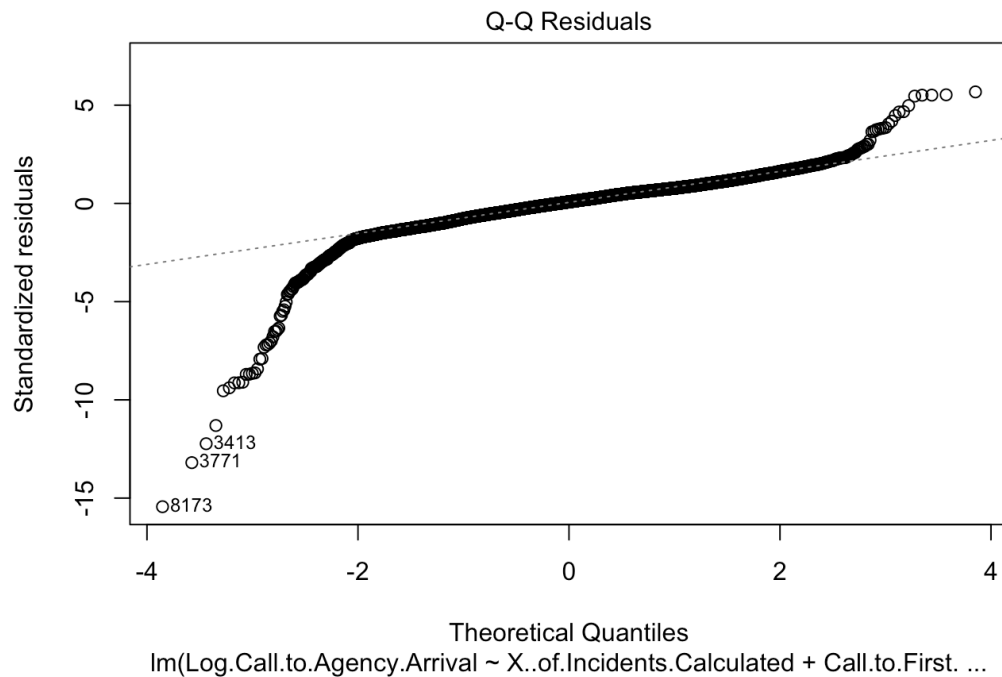
```

# Assumption checking
plot(responder_MLR, which=1)

```



```
plot(responder_MLR, which=2)
```



```
# install.packages("lmtest")  
library(lmtest)
```

Loading required package: zoo

Attaching package: 'zoo'

The following objects are masked from 'package:base':

as.Date, as.Date.numeric

```
# Breusch-Pagan (BP) test for heteroscedasticity  
bptest(responder_MLR)
```

studentized Breusch-Pagan test

data: responder_MLR
BP = 832.24, df = 26, p-value < 2.2e-16

```
# Kolmogorov-Smirnov (KS) Test for normality of residuals
```

```
ks.test(residuals(responder_MLR), "pnorm",
       mean = mean(residuals(responder_MLR)),
       sd = sd(residuals(responder_MLR)))
```

Asymptotic one-sample Kolmogorov-Smirnov test

```
data: residuals(responder_MLR)
D = 0.069477, p-value < 2.2e-16
alternative hypothesis: two-sided
```

```
#####
# Model did not pass; running model with      #
# log transformation                          #
#####
```

Log Transformed MLR

Next, an MLR model was created using the log transformed variables. Assumptions were also checked for this model, but still it didn't pass.

```
# Add categorical variables to the log transformed data
log_data_all <- log_data_transformed %>%
  mutate(Agency = explore$Agency,
         IncidentClassification =
           explore$IncidentClassification)

# MLR model with log transformed variables
responder_MLR_log <- lm(
  Log.Call.to.Agency.Arrival ~
    X.of.Incidents.Calculated + Call.to.First.Pickup +
    Call.to.PD.Calltaker.Handoff + Call.to.Agency.Dispatch +
    Call.to.First.Arrival..Multi.Agency.Incidents. +
    Median.Pickup +
    Median.Calltaker.Handoff + Median.Dispatch + Median.Travel
    +
    Median.Cumulative.First.Arrival...Multi.Agency.Incidents. +
    Average.Travel + Average.Dispatch +
    Average.Calltaker.Processing +
    Average.Pickup + as.factor(Agency) +
    as.factor(IncidentClassification),
  data = log_data_all
)

summary(responder_MLR_log)
```

Call:

```
lm(formula = Log.Call.to.Agency.Arrival ~
X..of.Incidents.Calculated +
  Call.to.First.Pickup + Call.to.PD.Calltaker.Handoff +
Call.to.Agency.Dispatch +
  Call.to.First.Arrival..Multi.Agency.Incidents. +
Median.Pickup +
  Median.Calltaker.Handoff + Median.Dispatch + Median.Travel
+
  Median.Cumulative.First.Arrival...Multi.Agency.Incidents. +
  Average.Travel + Average.Dispatch +
Average.Calltaker.Processing +
  Average.Pickup + as.factor(Agency) +
as.factor(IncidentClassification),
  data = log_data_all)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-0.56723	-0.01041	-0.00205	0.00745	0.76805

Coefficients:

	Estimate	Std. Error
(Intercept)	1.8673375	0.0428419
X..of.Incidents.Calculated	0.0011192	0.0003544
Call.to.First.Pickup	-0.0223095	0.0104979
Call.to.PD.Calltaker.Handoff	0.0054609	0.0066827
Call.to.Agency.Dispatch	0.3388075	0.0052584
Call.to.First.Arrival..Multi.Agency.Incidents.	0.2513026	0.0062186
Median.Pickup	0.0147232	0.0099833
Median.Calltaker.Handoff	-0.0080052	0.0027774
Median.Dispatch	0.0020703	0.0025910
Median.Travel	0.0808600	0.0048594
Median.Cumulative.First.Arrival...Multi.Agency.Incidents.	-0.0358326	0.0048725
Average.Travel	0.4728069	0.0055582
Average.Dispatch	-0.0638534	0.0049948

```

Average.Calltaker.Processing
0.0054586 0.0110500
Average.Pickup
0.0374039 0.1177065
as.factor(Agency)FDNY
-0.0605817 0.0043731
as.factor(Agency)NYPD
-0.0054365 0.0028263
as.factor(Agency)NYPD (Non-CIP)
0.0126385 0.0034471
as.factor(IncidentClassification)Critical
0.0094569 0.0033339
as.factor(IncidentClassification)Dispute
0.0056555 0.0018991
as.factor(IncidentClassification)Hazard
0.1967832 0.0053323
as.factor(IncidentClassification)Non-Critical
0.0214262 0.0034268
as.factor(IncidentClassification)Other
0.0030292 0.0024574
as.factor(IncidentClassification)Past Crime
-0.0055555 0.0024658
as.factor(IncidentClassification)Possible Crime
0.0043766 0.0018901
as.factor(IncidentClassification)Serious
0.0180552 0.0030450
as.factor(IncidentClassification)Vehicle Accident
-0.0090643 0.0023610

```

t

```

value Pr(>|t|)
(Intercept)
43.587 < 2e-16 ***
X..of.Incidents.Calculated
3.158 0.001595 **
Call.to.First.Pickup
-2.125 0.033604 *
Call.to.PD.Calltaker.Handoff
0.817 0.413851
Call.to.Agency.Dispatch
64.432 < 2e-16 ***
Call.to.First.Arrival..Multi.Agency.Incidents.
40.411 < 2e-16 ***
Median.Pickup
1.475 0.140308
Median.Calltaker.Handoff
-2.882 0.003958 **
Median.Dispatch
0.799 0.424289

```

```

Median.Travel
16.640 < 2e-16 ***
Median.Cumulative.First.Arrival...Multi.Agency.Incidents.
-7.354 2.10e-13 ***
Average.Travel
85.065 < 2e-16 ***
Average.Dispatch
-12.784 < 2e-16 ***
Average.Calltaker.Processing
0.494 0.621327
Average.Pickup
0.318 0.750665
as.factor(Agency)FDNY
-13.853 < 2e-16 ***
as.factor(Agency)NYPD
-1.924 0.054449 .
as.factor(Agency)NYPD (Non-CIP)
3.666 0.000247 ***
as.factor(IncidentClassification)Critical
2.837 0.004570 **
as.factor(IncidentClassification)Dispute
2.978 0.002909 **
as.factor(IncidentClassification)Hazard
36.904 < 2e-16 ***
as.factor(IncidentClassification)Non-Critical
6.253 4.23e-10 ***
as.factor(IncidentClassification)Other
1.233 0.217724
as.factor(IncidentClassification)Past Crime
-2.253 0.024286 *
as.factor(IncidentClassification)Possible Crime
2.316 0.020606 *
as.factor(IncidentClassification)Serious
5.929 3.16e-09 ***
as.factor(IncidentClassification)Vehicle Accident
-3.839 0.000124 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

```

Residual standard error: 0.03094 on 8541 degrees of freedom
Multiple R-squared:  0.9987,    Adjusted R-squared:  0.9987
F-statistic: 2.584e+05 on 26 and 8541 DF,  p-value: < 2.2e-16

```

```

# Assumption checking
bptest(responder_MLR_log)

```

studentized Breusch-Pagan test

```
data: responder_MLR_log  
BP = 1663.6, df = 26, p-value < 2.2e-16
```

```
ks.test(residuals(responder_MLR_log), "pnorm",  
        mean = mean(residuals(responder_MLR_log)),  
        sd = sd(residuals(responder_MLR_log)))
```

Asymptotic one-sample Kolmogorov-Smirnov test

```
data: residuals(responder_MLR_log)  
D = 0.19103, p-value < 2.2e-16  
alternative hypothesis: two-sided
```

```
#####  
# Model did not pass; running model with      #  
# BoxCox transformation                      #  
#####
```

BoxCox Transformed MLR

Since the models were not passing the KS test, a BoxCox transformation was attempted.

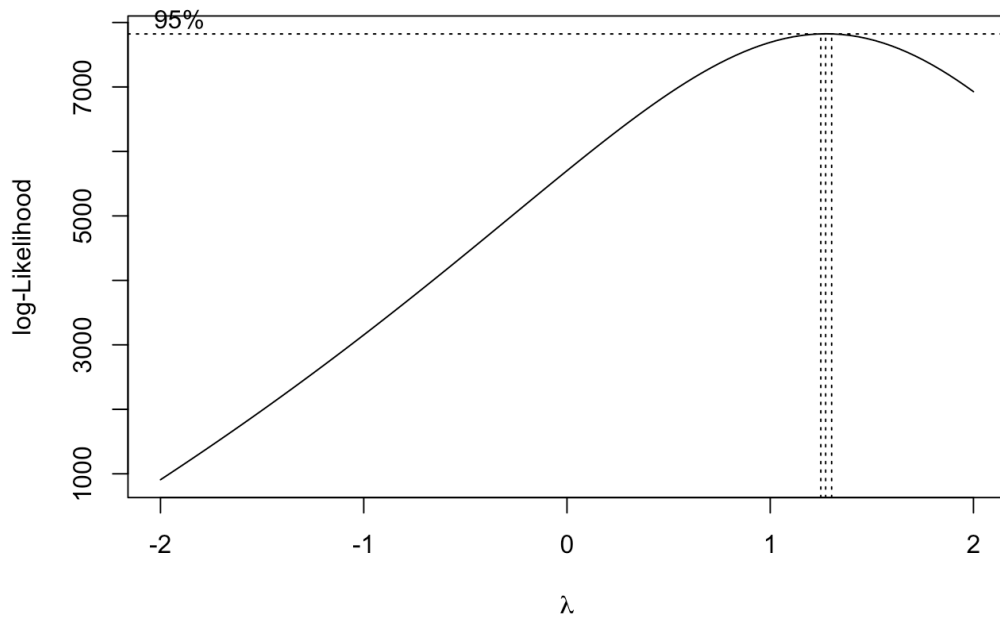
```
library(MASS)
```

Attaching package: 'MASS'

The following object is masked from 'package:dplyr':

select

```
# Apply BoxCox Transformation  
boxcox_results <- boxcox(responder_MLR_log, lambda = seq(-2,  
  2, by = 0.1))
```



```
# Identify the best lambda value
lambda_best <- boxcox_results$x[which.max(boxcox_results$y)]
lambda_best
```

```
[1] 1.272727
```

```
# Apply transformation to the response variable only
if (lambda_best > 0) {
  responder_clean$Transformed.Call.to.Agency.Arrival <-
    (responder_clean$Call.to.Agency.Arrival ^ lambda_best - 1)
    / lambda_best
} else if (lambda_best == 0) {
  responder_clean$Transformed.Call.to.Agency.Arrival <-
    log(responder_clean$Call.to.Agency.Arrival)
} else {
  responder_clean$Transformed.Call.to.Agency.Arrival <-
    -1 / (responder_clean$Call.to.Agency.Arrival ^
    lambda_best)
}

# MLR model with BoxCox-transformed response
responder_MLR_boxcox <- lm(
  Transformed.Call.to.Agency.Arrival ~
    X.of.Incidents.Calculated + Call.to.First.Pickup +
    Call.to.PD.Calltaker.Handoff + Call.to.Agency.Dispatch +
    Call.to.First.Arrival..Multi.Agency.Incidents. +
    Median.Pickup +
```



```

      Median.Calltaker.Handoff + Median.Dispatch + Median.Travel
      +
      Median.Cumulative.First.Arrival...Multi.Agency.Incidents. +
      Average.Travel + Average.Dispatch +
      Average.Calltaker.Processing +
      Average.Pickup + as.factor(Agency) +
      as.factor(IncidentClassification),
  data = responder_clean
)

summary(responder_MLR_boxcox)

```

Call:

```

lm(formula = Transformed.Call.to.Agency.Arrival ~
  X..of.Incidents.Calculated +
  Call.to.First.Pickup + Call.to.PD.Calltaker.Handoff +
  Call.to.Agency.Dispatch +
  Call.to.First.Arrival..Multi.Agency.Incidents. +
  Median.Pickup +
  Median.Calltaker.Handoff + Median.Dispatch + Median.Travel
+
  Median.Cumulative.First.Arrival...Multi.Agency.Incidents. +
  Average.Travel + Average.Dispatch +
  Average.Calltaker.Processing +
  Average.Pickup + as.factor(Agency) +
  as.factor(IncidentClassification),
  data = responder_clean)

```

Residuals:

	Min	1Q	Median	3Q	Max
	-7134.9	-317.3	-30.4	286.4	19586.0

Coefficients:

	Estimate	Std. Error
(Intercept)		
	-1.422e+03	9.293e+01
X..of.Incidents.Calculated		
	-1.519e-01	4.881e-03
Call.to.First.Pickup		
	9.996e+01	3.866e+01
Call.to.PD.Calltaker.Handoff		
	4.765e+00	1.577e+00
Call.to.Agency.Dispatch		
	8.217e-01	4.707e-01
Call.to.First.Arrival..Multi.Agency.Incidents.		
	-5.780e-01	1.153e-01

```

Median.Pickup
-1.222e+02  1.316e+01
Median.Calltaker.Handoff
-1.922e+00  4.407e-01
Median.Dispatch
-6.041e-01  1.294e-01
Median.Travel
1.584e+00  1.028e-01
Median.Cumulative.First.Arrival...Multi.Agency.Incidents.
9.580e-01  1.061e-01
Average.Travel
5.082e+02  6.460e+00
Average.Dispatch
4.563e+02  2.846e+01
Average.Calltaker.Processing
1.390e+02  8.949e+01
Average.Pickup
-5.436e+03  2.292e+03
as.factor(Agency)FDNY
-1.541e+02  5.930e+01
as.factor(Agency)NYPD
-9.817e+02  5.929e+01
as.factor(Agency)NYPD (Non-CIP)
-3.273e+03  6.861e+01
as.factor(IncidentClassification)Critical
7.015e+02  5.369e+01
as.factor(IncidentClassification)Dispute
5.072e+02  3.574e+01
as.factor(IncidentClassification)Hazard
-5.729e+02  8.671e+01
as.factor(IncidentClassification)Non-Critical
-5.036e+02  5.225e+01
as.factor(IncidentClassification)Other
2.026e+02  4.227e+01
as.factor(IncidentClassification)Past Crime
5.474e+02  4.752e+01
as.factor(IncidentClassification)Possible Crime
1.306e+03  3.640e+01
as.factor(IncidentClassification)Serious
2.535e+02  4.953e+01
as.factor(IncidentClassification)Vehicle Accident
-4.940e+02  4.507e+01

```

t

```

value Pr(>|t|)
(Intercept)
-15.297 < 2e-16 ***
X..of.Incidents.Calculated
-31.117 < 2e-16 ***

```

Call.to.First.Pickup
 2.586 0.00973 **
 Call.to.PD.Calltaker.Handoff
 3.021 0.00252 **
 Call.to.Agency.Dispatch
 1.746 0.08090 .
 Call.to.First.Arrival..Multi.Agency.Incidents.
 -5.012 5.46e-07 ***
 Median.Pickup
 -9.288 < 2e-16 ***
 Median.Calltaker.Handoff
 -4.362 1.30e-05 ***
 Median.Dispatch
 -4.668 3.08e-06 ***
 Median.Travel
 15.403 < 2e-16 ***
 Median.Cumulative.First.Arrival...Multi.Agency.Incidents.
 9.029 < 2e-16 ***
 Average.Travel
 78.669 < 2e-16 ***
 Average.Dispatch
 16.034 < 2e-16 ***
 Average.Calltaker.Processing
 1.553 0.12046
 Average.Pickup
 -2.372 0.01772 *
 as.factor(Agency)FDNY
 -2.598 0.00938 **
 as.factor(Agency)NYPD
 -16.557 < 2e-16 ***
 as.factor(Agency)NYPD (Non-CIP)
 -47.702 < 2e-16 ***
 as.factor(IncidentClassification)Critical
 13.065 < 2e-16 ***
 as.factor(IncidentClassification)Dispute
 14.192 < 2e-16 ***
 as.factor(IncidentClassification)Hazard
 -6.607 4.12e-11 ***
 as.factor(IncidentClassification)Non-Critical
 -9.639 < 2e-16 ***
 as.factor(IncidentClassification)Other
 4.792 1.67e-06 ***
 as.factor(IncidentClassification)Past Crime
 11.520 < 2e-16 ***
 as.factor(IncidentClassification)Possible Crime
 35.865 < 2e-16 ***
 as.factor(IncidentClassification)Serious
 5.118 3.14e-07 ***

```
as.factor(IncidentClassification)Vehicle Accident
```

```
-10.961 < 2e-16 ***
```

```
----
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Residual standard error: 663 on 10770 degrees of freedom
```

```
Multiple R-squared:  0.9969,    Adjusted R-squared:  0.9969
```

```
F-statistic: 1.321e+05 on 26 and 10770 DF,  p-value: < 2.2e-16
```

```
# Assumption checking  
bptest(responder_MLR_boxcox)
```

studentized Breusch-Pagan test

```
data: responder_MLR_boxcox
```

```
BP = 4393.3, df = 26, p-value < 2.2e-16
```

```
ks.test(residuals(responder_MLR_boxcox), "pnorm",  
        mean = mean(residuals(responder_MLR_boxcox)),  
        sd = sd(residuals(responder_MLR_boxcox)))
```

Asymptotic one-sample Kolmogorov-Smirnov test

```
data: residuals(responder_MLR_boxcox)
```

```
D = 0.08612, p-value < 2.2e-16
```

```
alternative hypothesis: two-sided
```

```
#####  
# Model did not pass; running model with      #  
# Winsorizing transformation                  #  
#####
```

Winsorized Model

After viewing the residual vs fitted plots and seeing the potential for outliers, a model was created using the winsorized data set.

```
# MLR Model after Winsorization  
responder_MLR_winz <- lm(Log.Call.to.Agency.Arrival ~  
    X.of.Incidents.Calculated+  
    Call.to.First.Pickup+  
    Call.to.PD.Calltaker.Handoff+  
    Call.to.Agency.Dispatch+  
    Call.to.First.Arrival + Multi.Agency.Incidents +
```

```

Call.to.First.Arrival..Multi.Agency.Incidents. +
  Median.Pickup+
  Median.Calltaker.Handoff+
  Median.Dispatch+
  Median.Travel+

Median.Cumulative.First.Arrival...Multi.Agency.Incident
  Average.Travel+
  Average.Dispatch+
  Average.Calltaker.Processing+
  Average.Pickup,
data = log_data_transformed[variables])
summary(responder_MLR_winz)

```

Call:

```

lm(formula = Log.Call.to.Agency.Arrival ~
X..of.Incidents.Calculated +
  Call.to.First.Pickup + Call.to.PD.Calltaker.Handoff +
Call.to.Agency.Dispatch +
  Call.to.First.Arrival..Multi.Agency.Incidents. +
Median.Pickup +
  Median.Calltaker.Handoff + Median.Dispatch + Median.Travel
+
  Median.Cumulative.First.Arrival...Multi.Agency.Incidents. +
  Average.Travel + Average.Dispatch +
Average.Calltaker.Processing +
  Average.Pickup, data = log_data_transformed[variables])

```

Residuals:

Min	1Q	Median	3Q	Max
-0.69950	-0.01267	-0.00306	0.00878	0.95617

Coefficients:

Estimate	Std. Error
(Intercept)	
2.3457178	0.0409290
X..of.Incidents.Calculated	
-0.0029849	0.0002742
Call.to.First.Pickup	
-0.0418024	0.0118106
Call.to.PD.Calltaker.Handoff	
-0.0094365	0.0068805
Call.to.Agency.Dispatch	
0.3722882	0.0055610
Call.to.First.Arrival..Multi.Agency.Incidents.	

0.2161034 0.0068330
Median.Pickup
0.0090107 0.0113686
Median.Calltaker.Handoff
0.0009752 0.0026268
Median.Dispatch
0.0346268 0.0012881
Median.Travel
0.1153135 0.0035027
Median.Cumulative.First.Arrival...Multi.Agency.Incidents.
-0.1736965 0.0040121
Average.Travel
0.5525034 0.0054279
Average.Dispatch
-0.0714160 0.0049635
Average.Calltaker.Processing
0.0298544 0.0114596
Average.Pickup
0.1608600 0.1347342

t

value Pr(>|t|)
(Intercept)
57.312 < 2e-16 ***
X..of.Incidents.Calculated
-10.884 < 2e-16 ***
Call.to.First.Pickup
-3.539 0.000403 ***
Call.to.PD.Calltaker.Handoff
-1.371 0.170256
Call.to.Agency.Dispatch
66.946 < 2e-16 ***
Call.to.First.Arrival..Multi.Agency.Incidents.
31.626 < 2e-16 ***
Median.Pickup
0.793 0.428035
Median.Calltaker.Handoff
0.371 0.710442
Median.Dispatch
26.881 < 2e-16 ***
Median.Travel
32.922 < 2e-16 ***
Median.Cumulative.First.Arrival...Multi.Agency.Incidents.
-43.294 < 2e-16 ***
Average.Travel
101.789 < 2e-16 ***
Average.Dispatch
-14.388 < 2e-16 ***
Average.Calltaker.Processing

```
2.605 0.009198 **
```

```
Average.Pickup
```

```
1.194 0.232548
```

```
----
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Residual standard error: 0.03577 on 8553 degrees of freedom
```

```
Multiple R-squared:  0.9983,    Adjusted R-squared:  0.9983
```

```
F-statistic: 3.59e+05 on 14 and 8553 DF,  p-value: < 2.2e-16
```

```
# Assumption checking  
bptest(responder_MLR_winz)
```

```
studentized Breusch-Pagan test
```

```
data: responder_MLR_winz
```

```
BP = 524.02, df = 14, p-value < 2.2e-16
```

```
ks.test(residuals(responder_MLR_winz), "pnorm",  
        mean = mean(residuals(responder_MLR_winz)),  
        sd = sd(residuals(responder_MLR_winz)))
```

```
Asymptotic one-sample Kolmogorov-Smirnov test
```

```
data: residuals(responder_MLR_winz)
```

```
D = 0.17796, p-value < 2.2e-16
```

```
alternative hypothesis: two-sided
```

```
#####  
# Model did not pass after winsorizing data; #  
# checking now for multicollinearity          #  
#####
```

Addressing Multicollinearity

At this point, the models continue to fail the assumption check so multicollinearity was examined.

```
# Assess multicollinearity  
eigen(cor(log_data_numeric))$values
```

```
[1] 8.541718268 2.294628992 1.570578708 0.983996963  
0.658161215 0.479857265
```

```
[7] 0.222128847 0.094482443 0.067645438 0.040437785  
0.027674697 0.010631309  
[13] 0.004235395 0.002648489 0.001174187
```

```
max(eigen(cor(log_data_numeric))$values)/min(eigen(cor(log_data_
```

```
[1] 7274.581
```

```
kappa(cor(log_data_numeric), exact = TRUE)
```

```
[1] 7274.581
```

```
model <- lm(Log.Call.to.Agency.Arrival~., data =  
            log_data_numeric)  
  
# install.packages("car")  
library(car)
```

Loading required package: carData

Attaching package: 'car'

The following object is masked from 'package:dplyr':

recode

```
# Assess variance inflation factor  
vif(model)
```

```
      X..of.Incidents.Calculated  
              1.219538  
      Call.to.First.Pickup  
              7.792503  
      Call.to.PD.Calltaker.Handoff  
              48.280018  
      Call.to.Agency.Dispatch  
              122.162548  
      Call.to.First.Arrival..Multi.Agency.Incidents.  
              231.424636  
      Median.Pickup  
              1.399948  
      Median.Calltaker.Handoff  
              14.709464  
      Median.Dispatch  
              16.811216
```


	Median.Travel	38.842579
Median.Cumulative.First.Arrival...Multi.Agency.Incidents.		54.017935
	Average.Travel	133.645325
	Average.Dispatch	142.890318
	Average.Calltaker.Processing	64.278414
	Average.Pickup	7.935850

```
mean(vif(model))
```

```
[1] 63.24359
```

```
# Remove variables with VIF greater than 55 and refit the
  model
vif_MLR <- lm(Log.Call.to.Agency.Arrival ~
              X..of.Incidents.Calculated +
              Call.to.First.Pickup +
              Median.Pickup + Average.Pickup+
              Median.Calltaker.Handoff+
              Median.Dispatch+
              Call.to.PD.Calltaker.Handoff+Median.Travel+
              Median.Cumulative.First.Arrival...Multi.Agency.Incident
              data = log_data_numeric)
summary(vif_MLR)
```

Call:

```
lm(formula = Log.Call.to.Agency.Arrival ~
    X..of.Incidents.Calculated +
      Call.to.First.Pickup + Median.Pickup + Average.Pickup +
      Median.Calltaker.Handoff +
      Median.Dispatch + Call.to.PD.Calltaker.Handoff +
      Median.Travel +
      Median.Cumulative.First.Arrival...Multi.Agency.Incidents.,
    data = log_data_numeric)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-0.92755	-0.09727	0.00158	0.09838	2.59152

Coefficients:

```

Estimate Std. Error
(Intercept)
0.108169    0.051150
X..of.Incidents.Calculated
-0.024276    0.001099
Call.to.First.Pickup
0.334612    0.024414
Median.Pickup
-0.368428    0.028416
Average.Pickup
-1.697117    0.230094
Median.Calltaker.Handoff
-0.157607    0.007728
Median.Dispatch
0.220678    0.003082
Call.to.PD.Calltaker.Handoff
0.415392    0.007528
Median.Travel
0.609357    0.004102
Median.Cumulative.First.Arrival...Multi.Agency.Incidents.
0.178129    0.005790

```

t

```

value Pr(>|t|)
(Intercept)
2.115    0.0345 *
X..of.Incidents.Calculated
-22.089  < 2e-16 ***
Call.to.First.Pickup
13.706  < 2e-16 ***
Median.Pickup
-12.966  < 2e-16 ***
Average.Pickup
-7.376  1.79e-13 ***
Median.Calltaker.Handoff
-20.395  < 2e-16 ***
Median.Dispatch
71.606  < 2e-16 ***
Call.to.PD.Calltaker.Handoff
55.181  < 2e-16 ***
Median.Travel
148.556  < 2e-16 ***
Median.Cumulative.First.Arrival...Multi.Agency.Incidents.
30.763  < 2e-16 ***

```

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.1528 on 8558 degrees of freedom
Multiple R-squared: 0.9693, Adjusted R-squared: 0.9693

F-statistic: 3.002e+04 on 9 and 8558 DF, p-value: < 2.2e-16

```
vif(vif_MLR)
```

```

X..of.Incidents.Calculated
1.122863
Call.to.First.Pickup
7.417878
Median.Pickup
1.349781
Average.Pickup
7.715460
Median.Calltaker.Handoff
7.418232
Median.Dispatch
5.584739
Call.to.PD.Calltaker.Handoff
4.794799
Median.Travel
3.834758
Median.Cumulative.First.Arrival...Multi.Agency.Incidents.
6.572942
```

```
# Assumption checking
bptest(vif_MLR)
```

studentized Breusch-Pagan test

```
data: vif_MLR
BP = 1012.5, df = 9, p-value < 2.2e-16
```

```
ks.test(residuals(vif_MLR), "pnorm",
        mean = mean(residuals(vif_MLR)),
        sd = sd(residuals(vif_MLR)))
```

Asymptotic one-sample Kolmogorov-Smirnov test

```
data: residuals(vif_MLR)
D = 0.046615, p-value < 2.2e-16
alternative hypothesis: two-sided
```

```
#####
# Model did not pass with log transformation;#
# attempting GLM with gamma distribution and #
```

```
# log link #
#####
```

Gamma Distributed GLM

After considering that the MLR is not able to work for the data, a new approach, a Generalized Linear Model with a Gamma distribution and log link, was taken.

```
# Gamma general linear model
glm_gamma <- glm(
  Call.to.Agency.Arrival ~ X..of.Incidents.Calculated +
    Call.to.First.Pickup + Call.to.PD.Calltaker.Handoff +
    Call.to.Agency.Dispatch +
    Median.Pickup +
    Median.Calltaker.Handoff + Median.Dispatch +
    Average.Calltaker.Processing +
    as.factor(Agency),
  data = train,
  family = Gamma(link = "log")
)

summary(glm_gamma)
```

```
Call:
glm(formula = Call.to.Agency.Arrival ~
X..of.Incidents.Calculated +
  Call.to.First.Pickup + Call.to.PD.Calltaker.Handoff +
Call.to.Agency.Dispatch +
  Median.Pickup + Median.Calltaker.Handoff + Median.Dispatch
+
  Average.Calltaker.Processing + as.factor(Agency), family =
Gamma(link = "log"),
  data = train)
```

Coefficients:

	Estimate	Std. Error	t value
Pr(> t)			
(Intercept)	5.894e+00	2.814e-02	209.460
< 2e-16 ***			
X..of.Incidents.Calculated	1.584e-05	1.818e-06	8.708
< 2e-16 ***			
Call.to.First.Pickup	2.855e-02	2.340e-03	12.199
< 2e-16 ***			
Call.to.PD.Calltaker.Handoff	3.061e-03	7.355e-04	4.162
3.18e-05 ***			

Call.to.Agency.Dispatch	1.056e-03	2.444e-05	43.216
< 2e-16 ***			
Median.Pickup	-1.765e-02	5.871e-03	-3.006
0.00266 **			
Median.Calltaker.Handoff	1.299e-03	1.632e-04	7.961
1.92e-15 ***			
Median.Dispatch	-1.524e-04	4.509e-05	-3.379
0.00073 ***			
Average.Calltaker.Processing	-1.142e-01	4.385e-02	-2.603
0.00924 **			
as.factor(Agency)FDNY	-3.342e-01	2.243e-02	-14.900
< 2e-16 ***			
as.factor(Agency)NYPD	-2.889e-02	2.206e-02	-1.310
0.19036			
as.factor(Agency)NYPD (Non-CIP)	7.459e-01	2.204e-02	33.845
< 2e-16 ***			

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for Gamma family taken to be 0.08113452)

Null deviance: 6178.83 on 8567 degrees of freedom
 Residual deviance: 690.77 on 8556 degrees of freedom
 AIC: 124067

Number of Fisher Scoring iterations: 8

```
# Assumption checking
bptest(glm_gamma)
```

studentized Breusch-Pagan test

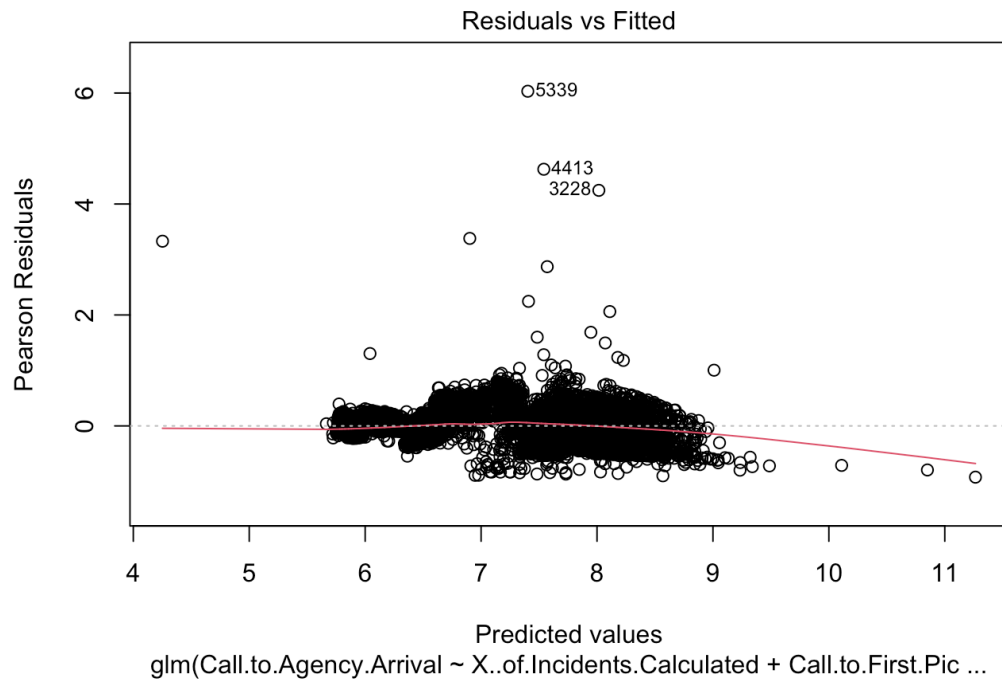
data: glm_gamma
 BP = 1167.9, df = 11, p-value < 2.2e-16

```
ks.test(residuals(glm_gamma), "pnorm",
        mean = mean(residuals(glm_gamma)),
        sd = sd(residuals(glm_gamma)))
```

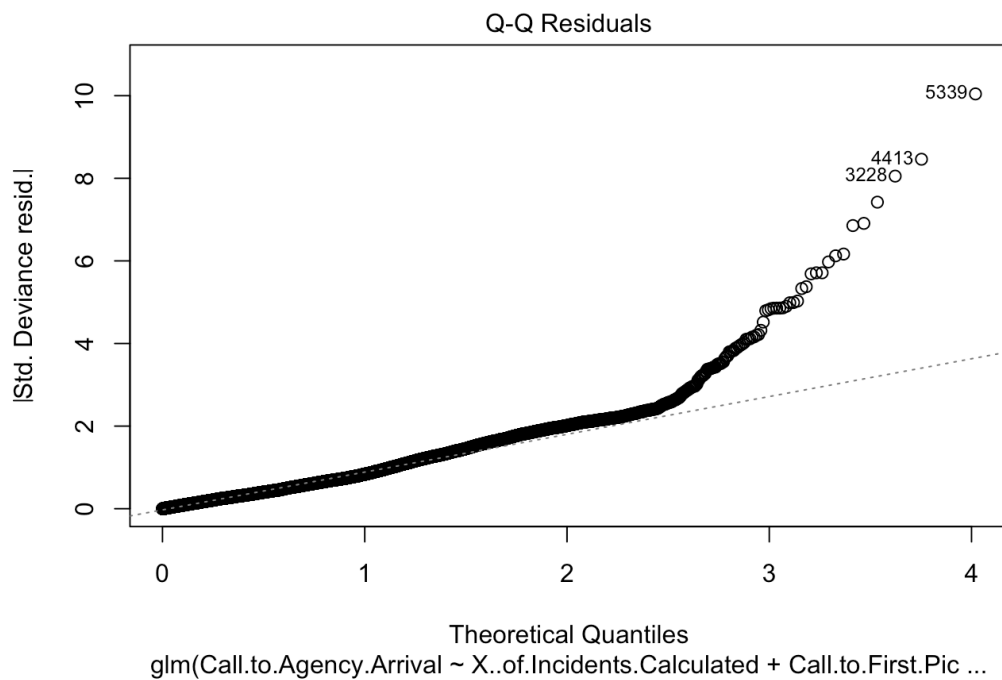
Asymptotic one-sample Kolmogorov-Smirnov test

data: residuals(glm_gamma)
 D = 0.078027, p-value < 2.2e-16
 alternative hypothesis: two-sided

```
# Residuals vs Fitted Values (Homoscedasticity & Linearity)
plot(glm_gamma, which = 1)
```



```
# Normal Q-Q Plot (Normality of residuals)
plot(glm_gamma, which = 2)
```



Ridge & LASSO Regression Model

To compare the GLM model created with other models, Ridge and LASSO regression models were generated.

```
# install.packages("glmnet")  
library(glmnet)
```

Loading required package: Matrix

Loaded glmnet 4.1-8

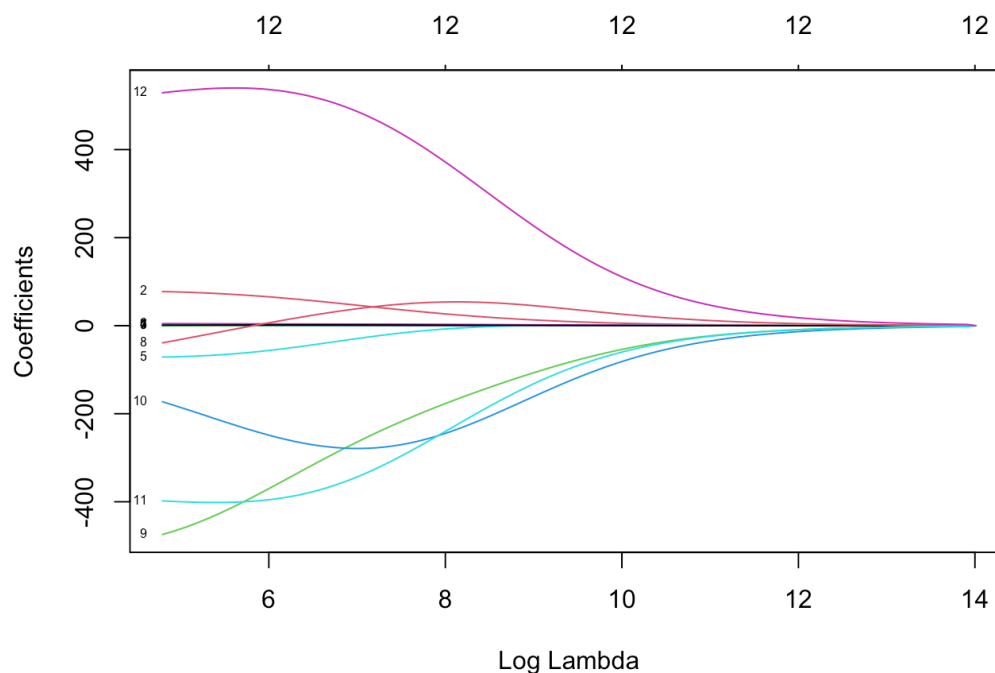
```
# RIDGE  
y <- train$Call.to.Agency.Arrival  
train_matrix <- as.matrix(train[, c(  
  "X..of.Incidents.Calculated",  
  "Call.to.First.Pickup",  
  "Call.to.PD.Calltaker.Handoff",  
  "Call.to.Agency.Dispatch",  
  "Median.Pickup",  
  "Median.Calltaker.Handoff",  
  "Median.Dispatch",  
  "Average.Calltaker.Processing"  
)])  
  
# Dummy code variables & split  
dummy_agency <- model.matrix(~ Agency - 1, data =  
  responder_clean)  
sample2 <- sample(c(TRUE, FALSE), nrow(dummy_agency),  
  replace=TRUE, prob=c(0.8,0.2))  
dummy_train <- dummy_agency[sample, ]  
dummy_test <- dummy_agency[!sample, ]  
  
# Combine dummy and numeric variables  
X <- cbind(train_matrix, dummy_train)  
  
# Fit a Ridge regression model  
ridge_model <- glmnet(X, y, alpha = 0, lambda = 0.1)  
coef(ridge_model)
```

13 x 1 sparse Matrix of class "dgCMatrix"

	s0
(Intercept)	202.3766405
X..of.Incidents.Calculated	0.0318465
Call.to.First.Pickup	76.4747732
Call.to.PD.Calltaker.Handoff	2.9284595
Call.to.Agency.Dispatch	2.3430100

Median.Pickup	-66.2422611
Median.Calltaker.Handoff	5.8063229
Median.Dispatch	0.1686729
Average.Calltaker.Processing	-333.0443909
AgencyEMS	-760.5028094
AgencyFDNY	-348.7812969
AgencyNYPD	-651.4691592
AgencyNYPD (Non-CIP)	191.8398962

```
# Perform cross-validation to select lambda
cv_ridge <- cv.glmnet(X, y, alpha = 0)
plot(cv_ridge$glmnet.fit, xvar = "lambda", label = TRUE)
```



```
best_lambda_ridge <- cv_ridge$lambda.min
ridge_model_best <- glmnet(X, y, alpha = 0, lambda =
  best_lambda_ridge)
coef(ridge_model_best)
```

13 x 1 sparse Matrix of class "dgCMatrix"

	s0
(Intercept)	33.0826334
X..of.Incidents.Calculated	0.0291279
Call.to.First.Pickup	77.6486806
Call.to.PD.Calltaker.Handoff	-0.3995282
Call.to.Agency.Dispatch	1.6957807
Median.Pickup	-71.2826644

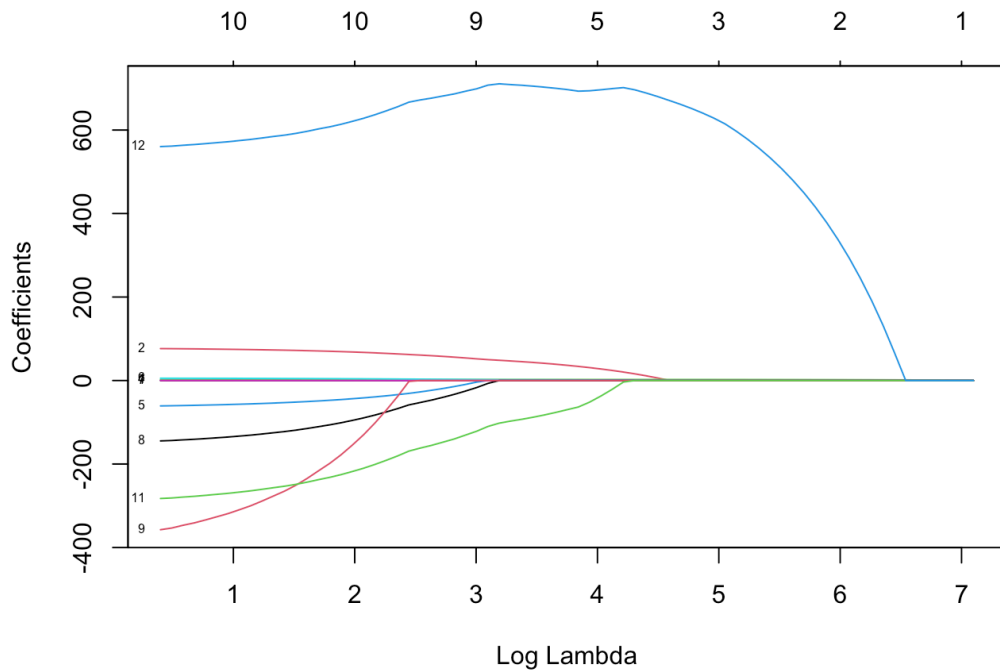
Median.Calltaker.Handoff	4.8171802
Median.Dispatch	1.1388331
Average.Calltaker.Processing	-37.5906524
AgencyEMS	-481.2799946
AgencyFDNY	-178.2502323
AgencyNYPD	-403.4344392
AgencyNYPD (Non-CIP)	524.7272807

```
# Fit a LASSO regression model
lasso_model <- glmnet(X, y, alpha = 1, lambda = 0.1)
coef(lasso_model)
```

13 x 1 sparse Matrix of class "dgCMatrix"

	s0
(Intercept)	159.26043935
X..of.Incidents.Calculated	0.03173972
Call.to.First.Pickup	76.73298962
Call.to.PD.Calltaker.Handoff	2.38475547
Call.to.Agency.Dispatch	2.34462486
Median.Pickup	-65.76905811
Median.Calltaker.Handoff	5.80307177
Median.Dispatch	0.16559688
Average.Calltaker.Processing	-300.40648008
AgencyEMS	-716.30439869
AgencyFDNY	-305.67294987
AgencyNYPD	-608.08285055
AgencyNYPD (Non-CIP)	234.98463291

```
# Perform cross-validation to select lambda
cv_lasso <- cv.glmnet(X, y, alpha = 1)
plot(cv_lasso$glmnet.fit, xvar = "lambda", label = TRUE)
```



```
best_lambda_lasso <- cv_lasso$lambda.min
lasso_model_best <- glmnet(X, y, alpha = 1, lambda =
  best_lambda_lasso)
coef(lasso_model_best)
```

13 x 1 sparse Matrix of class "dgCMatrix"

	s0
(Intercept)	-175.40305880
X..of.Incidents.Calculated	0.02349441
Call.to.First.Pickup	72.96089754
Call.to.PD.Calltaker.Handoff	.
Call.to.Agency.Dispatch	2.37140745
Median.Pickup	-53.32795943
Median.Calltaker.Handoff	4.98925467
Median.Dispatch	0.12552579
Average.Calltaker.Processing	-123.68433514
AgencyEMS	-276.89831978
AgencyFDNY	.
AgencyNYPD	-256.85347366
AgencyNYPD (Non-CIP)	585.36788654

Model Comparison

When considering these models, one way to consider which model is best is to compare the AIC and BIC values, and the RMSE values.

```
# install.packages("Metrics")
# install.packages("caret")
library(Metrics)
library(caret)
```

Loading required package: lattice

Attaching package: 'caret'

The following objects are masked from 'package:Metrics':

precision, recall

```
# Test sets
X_test2 <- as.matrix(test[, c(
  "X..of.Incidents.Calculated",
  "Call.to.First.Pickup",
  "Call.to.PD.Calltaker.Handoff",
  "Call.to.Agency.Dispatch",
  "Median.Pickup",
  "Median.Calltaker.Handoff",
  "Median.Dispatch",
  "Average.Calltaker.Processing"
)])

y_test <- test$Call.to.Agency.Arrival
dummy_inc_test <- cbind(X_test2, dummy_test)

# Predict with GLM on the test set
predictions <- predict(glm_gamma, newdata = test, type =
  "response")

# Make predictions with the best Ridge model
pred_ridge_test <- predict(ridge_model_best, newx =
  dummy_inc_test)

# Make predictions with the best Lasso model
pred_lasso_test <- predict(lasso_model_best, newx =
  dummy_inc_test)

# Calculate RMSE for all models
rmse_gamma <- rmse(test$Call.to.Agency.Arrival, predictions)
rmse_ridge <- rmse(y_test, pred_ridge_test)
rmse_lasso <- rmse(y_test, pred_lasso_test)

# Comparing model metrics
aic_gamma <- AIC(glm_gamma)
```

```

bic_gamma<- BIC(glm_gamma)

# Function to calculate AIC and BIC for glmnet models
calculate_AIC_BIC <- function(model, X, y) {
  # Predicted values from the model
  pred <- predict(model, newx = X)

  # Calculate residual sum of squares (RSS)
  rss <- sum((y - pred)^2)

  # Number of observations and number of coefficients
  n <- length(y)
  k <- sum(coef(model) != 0) # Number of non-zero coefficients

  # Calculate AIC and BIC
  aic <- n * log(rss / n) + 2 * k
  bic <- n * log(rss / n) + log(n) * k

  return(list(AIC = aic, BIC = bic))
}

# Calculate AIC and BIC for Ridge
ridge_metrics <- calculate_AIC_BIC(ridge_model_best, X, y)
print(ridge_metrics)

```

```

$AIC
[1] 111835.3

```

```

$BIC
[1] 111927

```

```

# Calculate AIC and BIC for LASSO
lasso_metrics <- calculate_AIC_BIC(lasso_model_best, X, y)
print(lasso_metrics)

```

```

$AIC
[1] 111661.6

```

```

$BIC
[1] 111739.3

```

```

# Create a data frame to hold the results
results_table <- data.frame(
  Model = c("GLM Gamma", "Ridge", "Lasso"),
  AIC = c(aic_gamma, ridge_metrics$AIC, lasso_metrics$AIC),
  BIC = c(bic_gamma, ridge_metrics$BIC, lasso_metrics$BIC),
  RMSE = c(rmse_gamma, rmse_ridge, rmse_lasso)
)

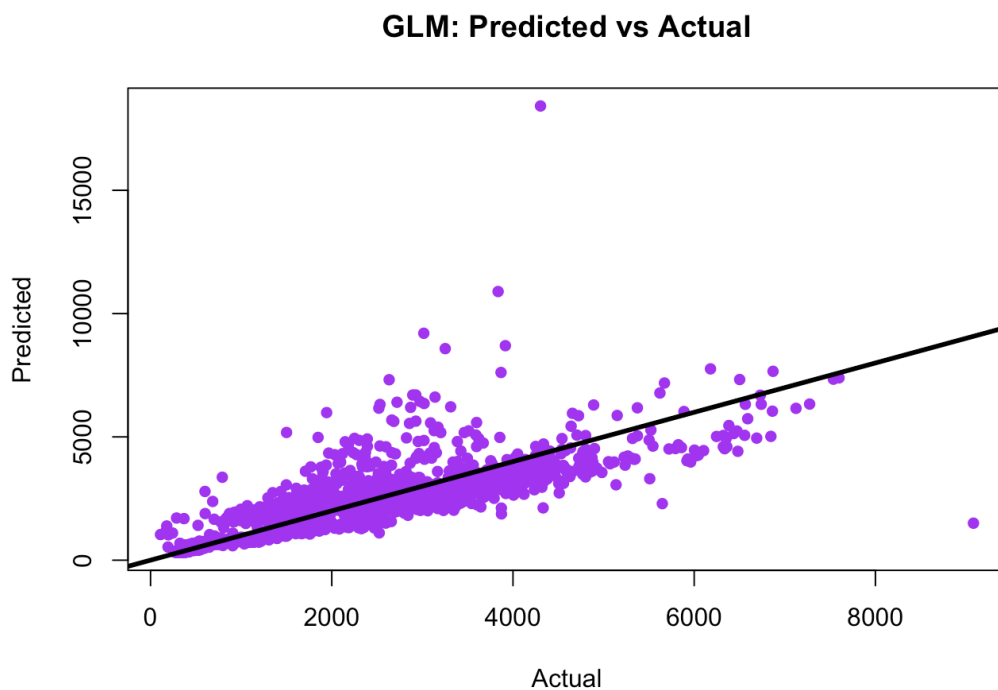
```

```
print(results_table)
```

	Model	AIC	BIC	RMSE
1	GLM Gamma	124066.7	124158.4	795.5191
2	Ridge	111835.3	111927.0	685.8036
3	Lasso	111661.6	111739.3	673.7035

Using the table above, it can be said that for this particular data set, the LASSO regression model is the best option. A visual representation of this can be seen from the graphs below as well.

```
# Plot Predicted vs Actual values for GLM
plot(test$Call.to.Agency.Arrival, predictions,
     main = "GLM: Predicted vs Actual",
     xlab = "Actual",
     ylab = "Predicted",
     col = "purple",
     pch = 16)
# Add identity line
abline(0, 1, col = "black", lwd=3)
```

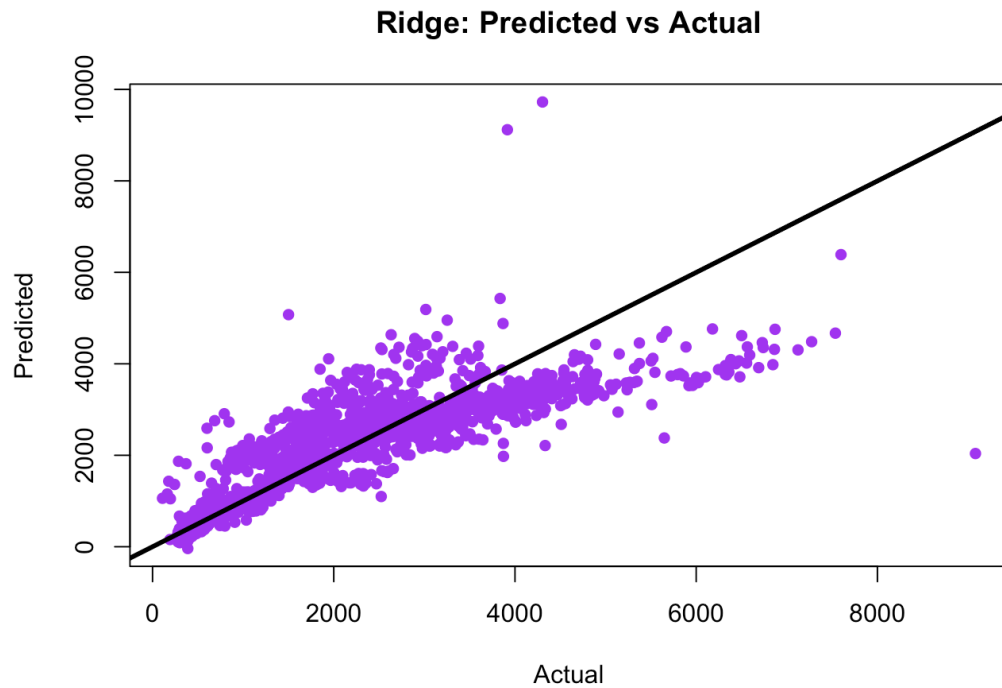


```
# Plot Predicted vs Actual values for Ridge
plot(
  y_test,
  pred_ridge_test,
  main = "Ridge: Predicted vs Actual",
```

```

xlab = "Actual",
ylab = "Predicted",
col = "purple",
pch=16)
# Add identity line
abline(0, 1, col = "black", lwd=3)

```



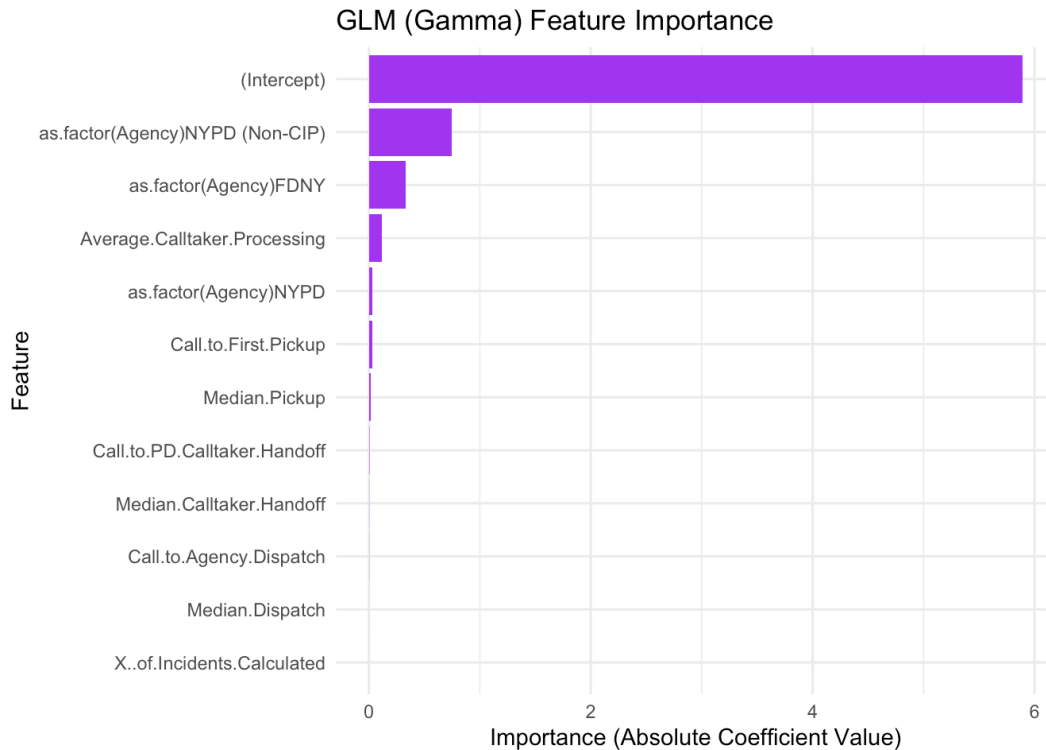
```

# Plot Predicted vs Actual values for LASSO
plot(
  y_test,
  pred_lasso_test,
  main = "LASSO: Predicted vs Actual",
  xlab = "Actual",
  ylab = "Predicted",
  col = "purple",
  pch=16)
# Add identity line
abline(0, 1, col = "black", lwd=3)

```



```
# Plot GLM feature importance
ggplot(importance_df, aes(x = reorder(feature, importance), y
  = importance)) +
  geom_bar(stat = "identity", fill = "purple") +
  coord_flip() +
  labs(
    title = "GLM (Gamma) Feature Importance",
    x = "Feature",
    y = "Importance (Absolute Coefficient Value)"
  ) +
  theme_minimal()
```



```
# Feature Importance for LASSO regression
# Get coefficients
coef_lasso <- coef(lasso_model_best)

# Convert to tidy data frame
coef_df <- as.data.frame(as.matrix(coef_lasso))
coef_df$Feature <- rownames(coef_df)
colnames(coef_df)[1] <- "Coefficient"

# Remove intercept and zero coefficients
coef_df <- coef_df[coef_df$Feature != "(Intercept)" &
  coef_df$Coefficient != 0, ]

# Plot LASSO feature importance
ggplot(coef_df,
```



```

aes(x = reorder(Feature, abs(Coefficient)),
     y = abs(Coefficient))) +
geom_bar(stat = "identity", fill = "purple") +
coord_flip() +
labs(title = "LASSO Feature Importance",
     x = "Feature",
     y = "Importance (Absolute Coefficient Value)") +
theme_minimal()

```

